

Interpreting reconstruction scores in the 2026 ICASSP Hyper-Object Challenge

Abstract

The 2026 ICASSP Hyper-Object Challenge [1] evaluates hyperspectral reconstruction using the composite Spectral-Spatial-Color score (SSC). We examine what this score communicates about reconstruction errors by scoring deliberately modified ground-truth cubes against their originals, separately within regions of interest containing objects and in the surrounding regions. Gaussian noise with standard deviation 0.003 gives an SSC score of 0.904 in the region of interest but only 0.019 in one surrounding region (higher SSC score is better). We also find that, because of the non-linear nature of the score calculation, substantial changes in individual reconstruction errors can leave their contribution to SSC unchanged. Experiments with MST++ [2], MSFN [3] and SPECAT [4] show that restricting training to the region of interest increases its SSC score by up to 3.8 percentage points compared with training that includes surrounding pixels. We also show that increasing MST++ [2] training from 100 to 500 epochs improves region-of-interest SSC scores by up to 6.1 percentage points. An MSFN [3]–SPECAT [4] ensemble improves these scores by up to 1.4 percentage points over the better individual model and improves all submetrics.

1 Introduction

The 2026 ICASSP Hyper-Object Challenge: Track 1 on Kaggle asks models to reconstruct hyperspectral reflectance images of everyday objects from camera-like inputs [1]. A model receives a single-channel color mosaic and predicts 61 wavelength values at every pixel. The challenge reports reconstruction quality through the Spectral-Spatial-Color score, SSC, which combines six submetrics.

In this paper, we examine what this score communicates about prediction errors. We apply known changes to ground-truth cubes and measure how the submetrics and SSC respond across image regions. We then compare MST++, MSFN and SPECAT under different training policies, examine the effects of longer MST++ training, and evaluate an ensemble of MSFN and SPECAT predictions. These experiments let us examine which errors improve when a model’s score improves, and where in the image those improvements occur.

Our experiments use a local scoring formula recovered from the organizers’ Python bytecode. The overview paper and recovered code give different scoring definitions. Section 2.3 specifies the calculation used here; Appendix E documents its differences from the deployed Kaggle evaluator and the controlled submissions used to examine them.

2 The Hyper-Object Challenge

2.1 Reconstructing spectra from a mosaic

The ICASSP Hyper-Object Challenge studies reconstruction of hyperspectral images of everyday objects from lower-cost, camera-like inputs [5]. Track 1 asks a model to reconstruct spectra from a raw mosaic. A model receives one simulated color response per pixel and must predict the object’s spectral reflectance as well as that of the surrounding scene.

Each target is a measured reflectance cube with 1024×1024 pixels and 61 wavelength bands, spaced every 10 nm from 400 to 1000 nm. A cube can be read as 61 images, one per wavelength, or as a 61-value spectrum at each pixel. The input is simulated from this target: the organizers render it to linear RGB, then retain one red, green or blue response at each location according to a repeating RGGGBayer pattern [6]. This produces a single-channel 1024×1024 mosaic. Input and target arrays contain floating-point values in $[0, 1]$.

The simulated RGB input covers the visible range, whereas the target also includes near-infrared bands. The organizers describe the input as containing no direct near-infrared observations [5]. Reconstruction of the 710–1000 nm target bands therefore depends on relationships learned from the training pairs. Even within the visible range, each pixel supplies only one color response for reconstructing a full spectrum.

The released images show apples, books, drinking glasses and household objects such as packages, bottles and cans, photographed with a white sheet, dark table and surrounding background. Figure 1 illustrates the input/output task and examples from the released targets. Each target cube includes the entire scene.

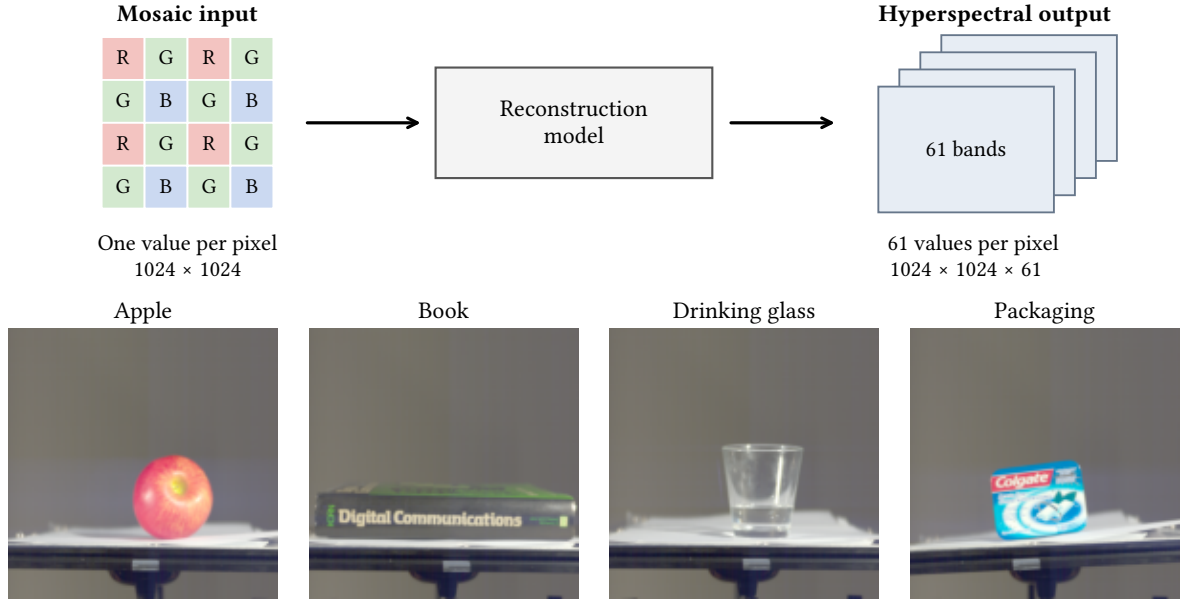


Figure 1: The Track 1 task and example scenes. Top: a schematic of RRGB sampling and the required output. The colored cells label the sampled response at each position. Bottom: display-brightened RGB renderings of four dataset images.

2.2 Original partitions and reported results

The organizers’ overview paper describes 167 training samples and three evaluation sets: eleven public, four private and four hidden samples. Ground truth is available for the training and public sets [5].

The final assessment used hidden data and additional unseen samples. In the overview’s final Track 1 ranking, Capybara placed first with an SSC of 0.693217, followed by GISP@HEU at 0.688150 and ACVLAB at 0.636549 [5].

Our Kaggle probes use only the four private mosaics, so their scores should not be confused with these final challenge results. Our local experiments also use different evaluation conditions: a separate data split, described in Section 3, and the local scoring calculation defined in Section 2.3. Kaggle and final challenge scores are therefore not directly comparable to our local SSC scores.

2.3 SSC and its submetrics

The Spectral-Spatial-Color score (SSC) combines spectral, spatial and color agreement on a stated scale from zero to one, with larger values indicating closer agreement. The organizers describe the score in their overview paper, **The Hyper-Object Challenge: Inferring Beyond the Visible in Low-Cost Hyperspectral Imaging** [5], and their hyper-object/2026-ICASSP-SPGC repository [7].

For our study, we use the combination recovered from the organizers’ retained code:

$$\text{SSC} = S_{\text{spec}}^{0.5} S_{\text{spat}}^{0.35} S_{\text{col}}^{0.15}. \quad (1)$$

SSC is the weighted geometric mean of the spectral, spatial and color similarities, denoted S_{spec} , S_{spat} and S_{col} .

The function $\text{clip}_{[a,b]}(x)$ limits x to the interval $[a, b]$: values below a become a , values above b become b , and values within the interval are unchanged.

For a measured error m and a positive scale τ in the same units, we define an exponential similarity:

$$E_{\tau}(m) = \text{clip}_{[10^{-6}, 1]}(e^{-\frac{m}{\tau}}). \quad (2)$$

SAM, the spectral angle mapper, averages the angle between predicted and target spectra in degrees. **SID**, spectral information divergence, compares their normalized band proportions. **ERGAS** combines each band’s root-mean-square error relative to the mean target reflectance. Lower values indicate closer agreement for all three. Their transformed similarities form a geometric mean:

$$S_{\text{spec}} = (E_5(\text{SAM})E_{0.02}(\text{SID})E_3(\text{ERGAS}))^{\frac{1}{3}}. \quad (3)$$

The scales are 5 degrees for SAM, 0.02 for SID and 3 for ERGAS.

The appearance metrics use RGB rendered from the visible 400–700 nm bands. **PSNR** converts mean squared RGB error to decibels, with higher values indicating closer agreement. Its similarity score S_{PSNR} rescales 20–50 dB to zero–one and clips values outside that interval:

$$S_{\text{PSNR}} = \text{clip}_{[0,1]} \left(\frac{\text{PSNR} - 20}{30} \right). \quad (4)$$

SSIM measures similarity in local RGB neighborhoods, with higher values better. The spatial group averages SSIM and the rescaled PSNR:

$$S_{\text{spat}} = \frac{S_{\text{PSNR}} + \text{SSIM}}{2}. \quad (5)$$

CIEDE2000, written ΔE_{00} , measures color difference, with lower values better. The color group applies the error-to-similarity transformation with scale 3:

$$S_{\text{col}} = E_3(\Delta E_{00}). \quad (6)$$

Appendix B specifies the raw metric definitions and rendering; Appendix E.2 gives the full combination conventions. Once a transformed similarity reaches its lower bound, further increases in the corresponding error leave its contribution to SSC unchanged. Improvements also remain hidden until the error falls below the clipping threshold.

The overview paper describes a weighted arithmetic combination of the three groups and PSNR over the hyperspectral cube [5]. The repository’s Python bytecode instead gives the geometric combination above, with PSNR and SSIM calculated on rendered RGB [7]. The accompanying source contains a different “toy” scorer. We use the recovered combination for our local experiments and compute the metrics within each evaluated region. Despite our extended efforts, we believe our local SSC calculation does not fully reproduce the deployed Kaggle scorer. Appendix E presents the methods we used to attempt to reverse-engineer the scorer, Kaggle scores from our probes, and known remaining divergences.

3 Dataset examination

3.1 Scene coverage and local split

The supplied files are organized into train, test-public and test-private. Released targets accompany the training and public pairs, so the public set can be used for validation. Targets for the four private mosaics are withheld; Kaggle scores submitted predictions for those inputs.

The public set contains nine apple images and two glass images, with no books or other household objects. The four private inputs show an apple, a book, a drinking glass and a bottle. Figure 2 compares their scene types with our local validation allocation. The separate hidden and additional final-assessment images are outside this examination.

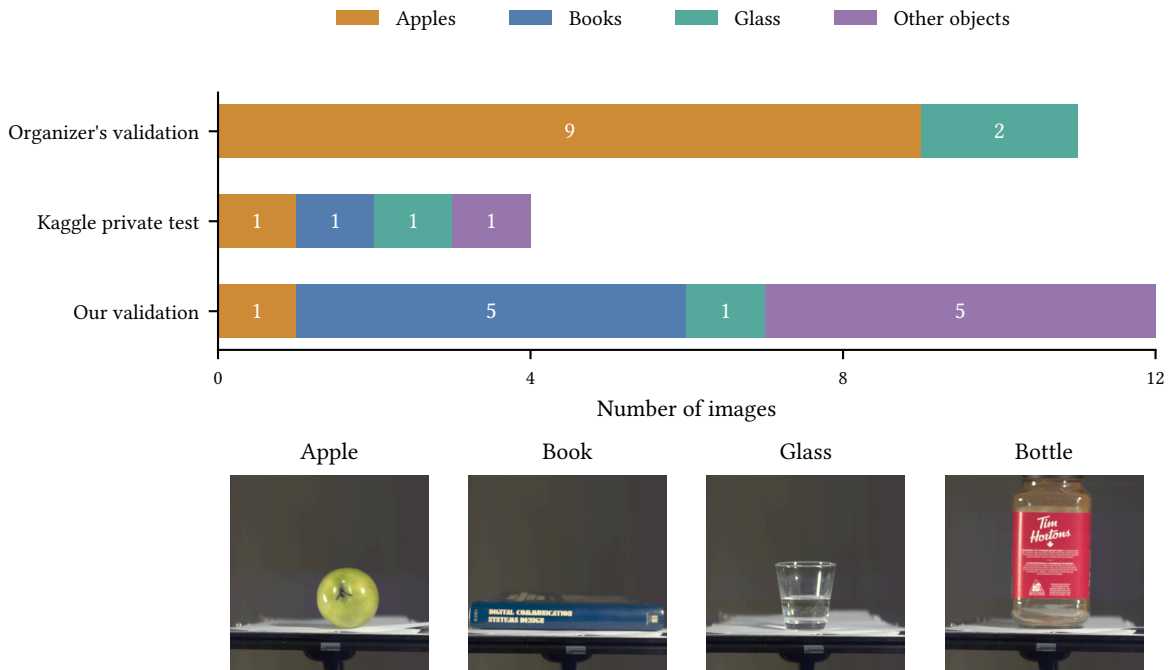


Figure 2: Scene-type counts in the Organizers’ set, the Kaggle leaderboard private set, and our local validation set. Bottom: demosaiced RGB previews of the four private Kaggle inputs, brightened for display.

The dataset shows repeated views of similar apples and drinking glasses, with changes in orientation, occupied image area and brightness. Appendix A.2 shows all 178 released target previews, grouped by category with partition and holdout labels, so the repeated views can be inspected directly.

Public validation omits books and other objects that are present in the supplied training data. To examine those scene types too, we recombined the available labeled pairs into a local split of 165 training and twelve validation images, with one image excluded. We did not reserve an additional local test set; the Kaggle private evaluation remains separate.

Our twelve validation images come from the original train partition: one apple, five books, one glass and five household objects. The training set contains the remaining 154 usable train pairs and all eleven test-public pairs. Appendix A records the identifiers and categories. Figure 11 shows all twelve validation images.

3.2 Masks and regions

To distinguish reconstruction of the objects from reconstruction of their surroundings, we report full-frame metrics and evaluate three disjoint regions separately. **Object + sheet** includes the object and the white paper beneath it. **Table** is an 80-pixel-high band below the fitted lower sheet boundary. **Background** is the remainder. The masks are derived algorithmically and are constant across training

and validation runs. Appendix A.1 records how the masks were constructed. Section 4 examines how these regions exhibit different sensitivities to synthetic noise.

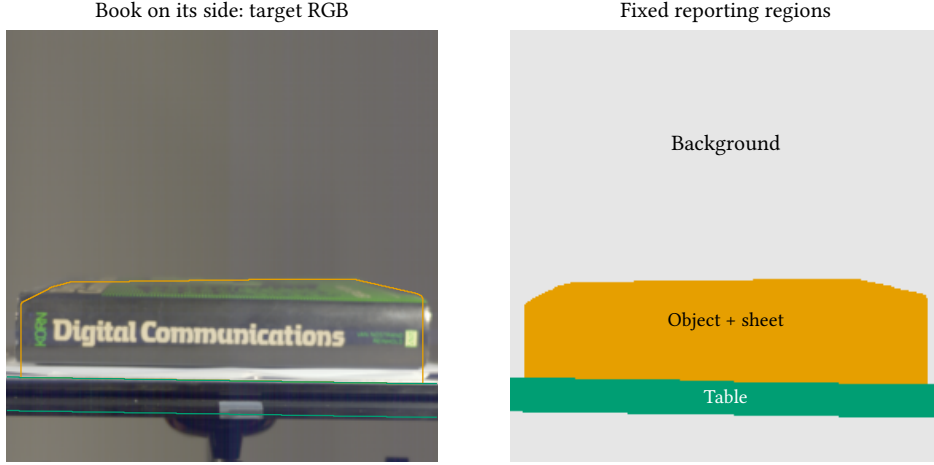


Figure 3: Reporting regions for the book lying on its side. The target RGB rendering and corresponding masks show the white sheet included with the object, the narrow table band below it, and the remaining background.

4 Metric sensitivity analysis

We start with a ground-truth cube, apply a controlled change, and score the changed cube against the unchanged original. This lets us relate the score to a known error. We first add independent Gaussian noise to every band value, then clip negative values to zero. Its standard deviation σ is measured in reflectance units. We calculate metrics and SSC for each image and region, then average the twelve images equally.

Small Gaussian disturbances produce large regional differences in SSC. At $\sigma=0.003$, mean object + sheet SSC remains 0.904, whereas full-frame SSC is 0.407 and table SSC is 0.019. At $\sigma=0.01$, these values become 0.657, 0.040 and 0.002. Figure 4 shows the sweep and the individual-image curves.

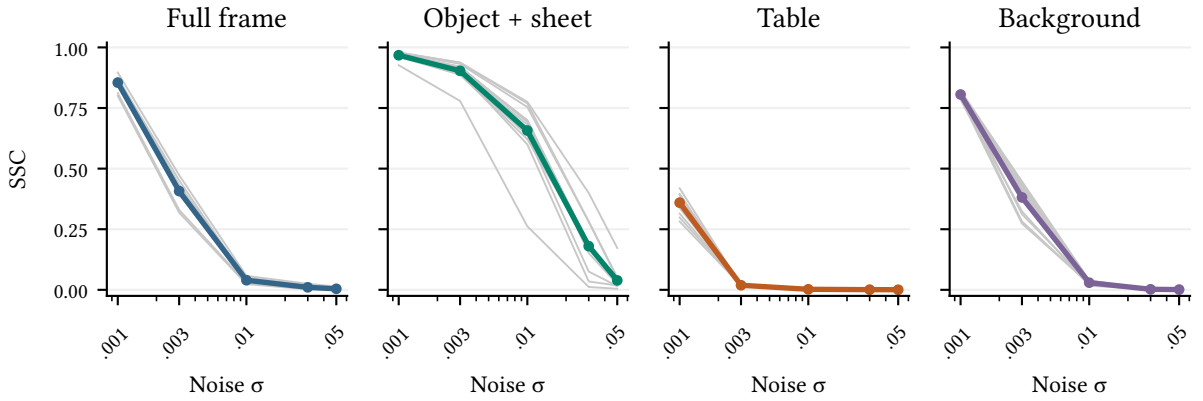


Figure 4: SSC after additive Gaussian noise and negative-value clipping. Thin gray curves show our twelve validation images. Colored curves show their equal-weight mean. All four panels use the same SSC scale. The horizontal axis is logarithmic.

Before clipping, the noise has the same standard deviation throughout the image. It produces a larger relative disturbance in low-reflectance spectra. SAM measures direction, SID compares normalized band proportions, and ERGAS normalizes bandwise error by the region's target mean. The raw metrics make the regional contrast explicit: at $\sigma=0.01$, mean SAM is 2.53° on object + sheet and 30.26° on table; SID is 0.0094 and 3.5452 respectively. RGB appearance metrics also change, as Figure 5 shows.

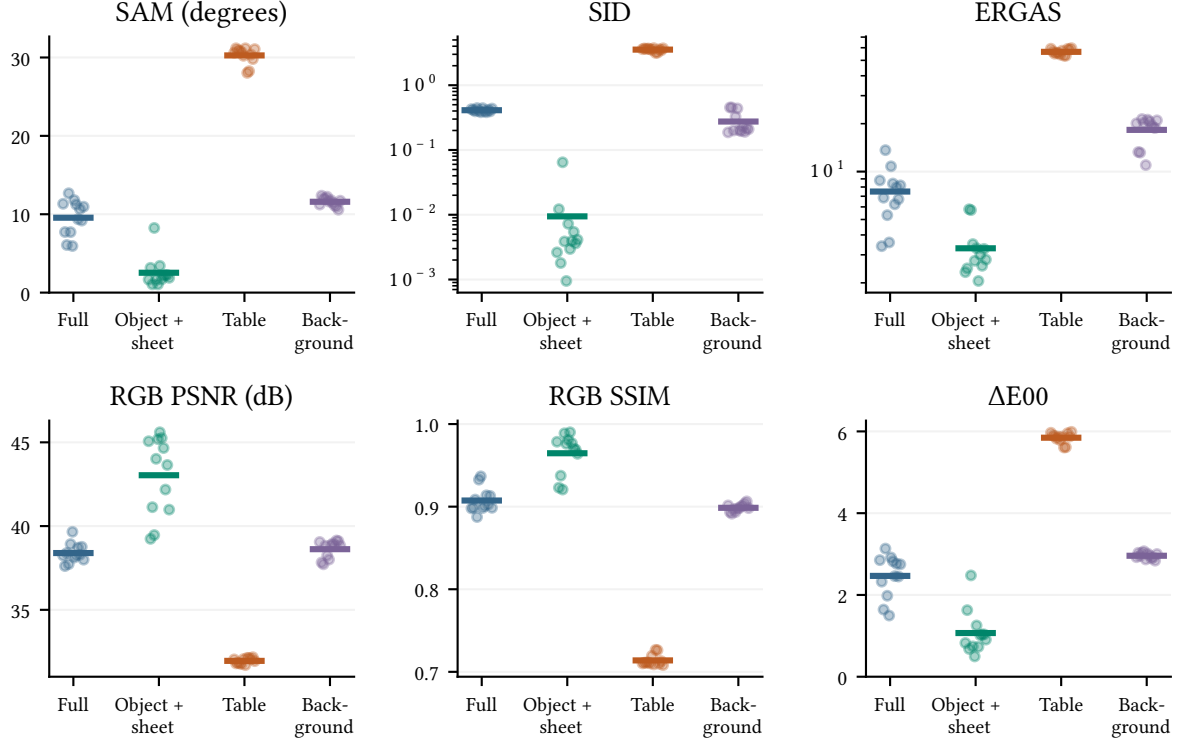


Figure 5: The six raw metrics after $\sigma=0.01$ Gaussian noise and negative clipping. Dots show individual validation images out of a total of 12; horizontal marks show the twelve-image means. SID and ERGAS use logarithmic scales. Lower is better for SAM, SID, ERGAS and ΔE_{00} ; higher is better for PSNR and SSIM.

Negative-value clipping also matters. It creates zeros and changes the noise distribution most strongly where the original signal is small. The observed response combines the effects of clipping with the metrics' sensitivity to low reflectance. The separate brightness-group analysis in Appendix C.4.1 shows the same dependence within individual images.

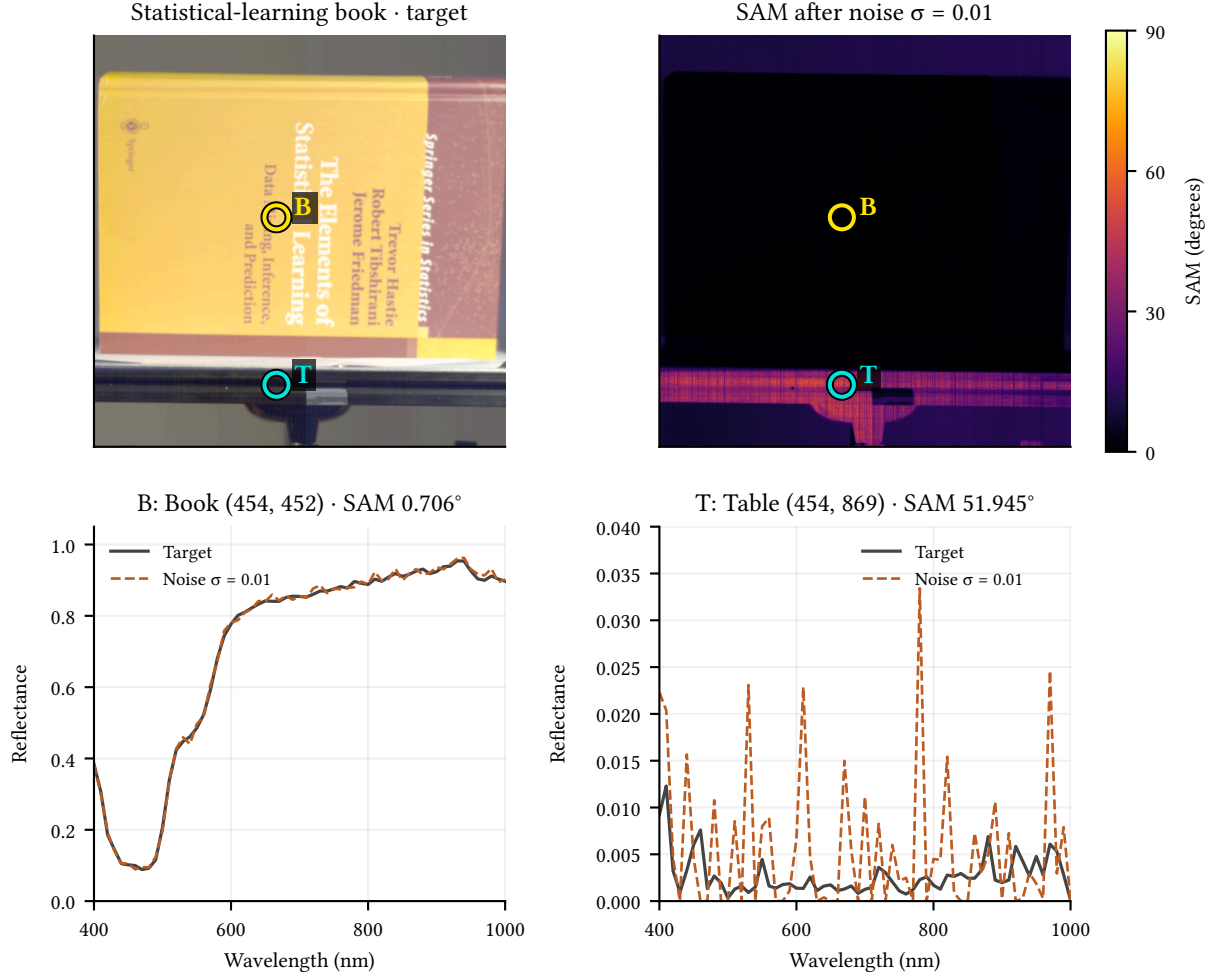


Figure 6: Gaussian noise at two points in the statistical-learning book scene, with $\sigma=0.01$. Top: target RGB and the full-frame SAM (loss) map. Bottom: target and target + noise spectra at book point B, and table point T. SAM is 0.706° at B and 51.945° at T. Negative values are clipped to zero.

At the table point in Figure 6, the original spectrum lies close to zero. The same nominal disturbance that makes small changes to the book spectrum substantially changes the table spectrum’s direction. Clipping sets 20 of its 61 perturbed bands to zero, versus none at the book point.

Spatial averaging provides another way to relate an SSC value to a known change in the reconstruction. Averaging each 2×2 block of ground truth and bilinearly resizing it to the original dimensions gives mean SSC 0.502 for the full frame and 0.671 for object + sheet. Appendix C.1.1 shows the pooling and pixel-replacement results in detail. The remaining perturbations and complete measurements are in Appendix C.

5 Metric behavior in trained reconstructions

The controlled perturbations show how known changes affect the metrics. We now examine trained models under different training policies and budgets, and after averaging their predictions. We evaluate each region separately to see where the scores improve and which errors change.

5.1 Models and training setup

We compare MST++, MSFN and SPECAT using patches from the same 165 training images and full-image evaluation on the same twelve validation images. Our reproduction of the organizers’ Raw2HSI baseline scored 0.27299 on Kaggle. Appendix D records that reference and the Raw2HSI baseline experiments.

Raw-input MST++ processes the mosaic through three successive spectral transformers. Each combines attention across feature channels with spatial convolutions and an encoder-decoder [2]. Packed-input MST++ separates each 2×2 RGGB block into four channels, then restores full resolution through learned upsampling in both the processed-feature path and the shallow-feature skip connection [8]. Its four 64×64 input planes contain the same measurements as a 128×128 raw patch.

MSFN and SPECAT use Menon demosaicing to estimate missing RGB values on the complete mosaic, then extract crops. MSFN links spectral and spatial encoder-decoders [9]. SPECAT applies spatial attention before spectral attention within each block, with a separate guidance branch learned from the input RGB [10]. Both follow the model families used by SATE [11]. All configurations use base width 64, meaning channel width 64, which we trim to 61 for final prediction (this helps the performance of CUDA training kernels).

Each update samples four 128×128 patches from four distinct training images. A **reference epoch** is 165 updates; 100 reference epochs therefore mean 16,500 updates and 66,000 patches. The **masked policy** samples crops with at least 50% object + sheet coverage and applies loss only to those pixels. The **unmasked policy** samples all allowed crop origins and applies loss to every pixel.

Training uses AdamW and clamps predictions to $[0, 1]$ before calculating the common loss, $L1 + 0.1 \text{ SAM}$, with SAM in radians. Two SPECAT variants replace this objective with SATE’s Integral Spectral, or IS, loss: $10L1 + \frac{\text{SAM}_{\text{rad}}}{3} + \text{NSE}$, where NSE is absolute spectral error divided by target spectral strength. Appendix F.3 defines its units and reductions.

For validation, we use the weights saved at the end of each run and predict complete images. The main comparisons clip predictions to $[0, 1]$ and leave the targets unchanged. This prediction clipping precedes metric calculation; the lower bound on transformed similarities in Section 2.3 is a separate scoring operation. We use the same masks and evaluator throughout, calculate each image’s metrics and SSC, then average images equally. Appendix F records the remaining settings and sources.

5.2 Masked versus unmasked training

Across all five configurations trained for 100 reference epochs, masked training gives higher object + sheet SSC, while unmasked training gives higher full-frame, table and background SSC (Figure 7). Which training policy scores better therefore depends on the evaluation region.

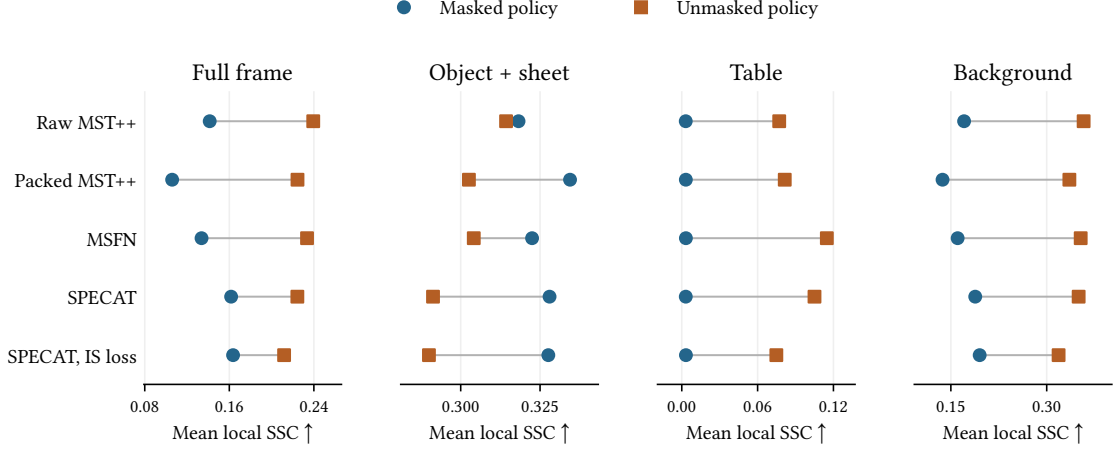


Figure 7: The ten models after 100 reference epochs, evaluated on twelve images with predictions clipped to $[0, 1]$. Blue circles are masked training and orange squares are unmasked training.

Within either policy, differences between architectures are smaller than the changes caused by including the surroundings in evaluation. The four masked models trained with the common loss span object + sheet SSC 0.318–0.334, and their unmasked counterparts span 0.291–0.314.

A higher SSC need not coincide with lower spectral error. For raw-input MST++, masked training raises object + sheet SSC from 0.314 to 0.318 while mean SAM, SID and ERGAS worsen. All three rendered-appearance metrics improve. The other three configurations trained with the common loss improve all three spectral means under masking.

5.3 What changes with longer training?

We train raw-input and packed-input MST++ under both policies for 500 reference epochs, or 82,500 updates. The 100-epoch runs use their own 100 epoch schedules, and are not base models for the 500 epoch runs. The comparison therefore changes both the number of updates and the learning-rate trajectory.

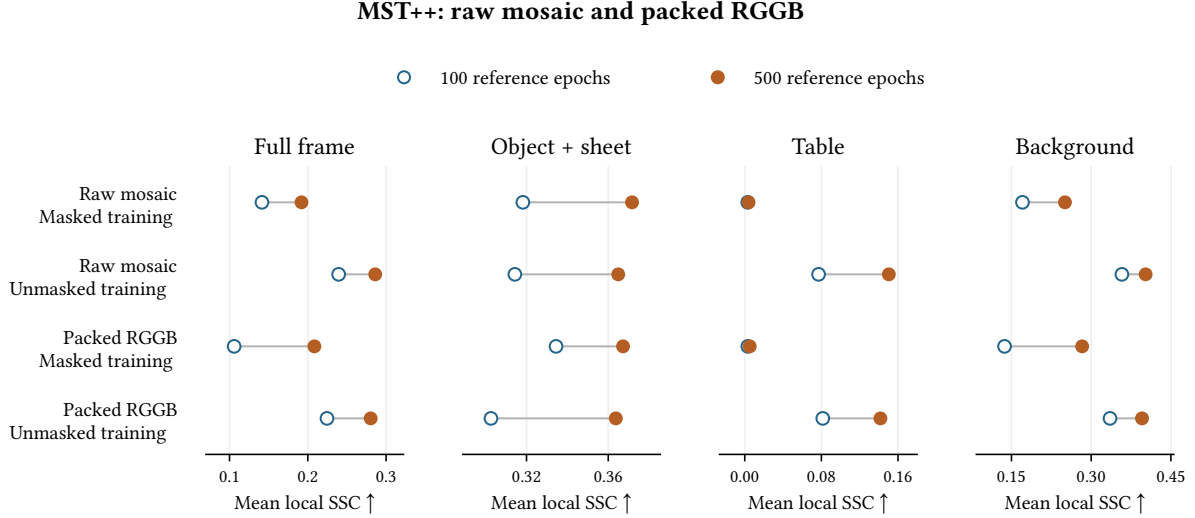


Figure 8: Raw-input and packed-input MST++ after 100 and 500 reference epochs. Open blue circles show 100 epochs; filled orange circles show 500. Values are twelve-image mean SSC with predictions clipped to $[0, 1]$.

All four 500-reference-epoch runs improve mean object + sheet SSC over their 100-reference-epoch counterparts, reaching 0.364–0.372. The masked models change order: packed input leads at 100 epochs, but raw input has slightly higher object + sheet SSC at 500, 0.3716 versus 0.3672. Figure 9 shows how that change differs across metrics and regions.

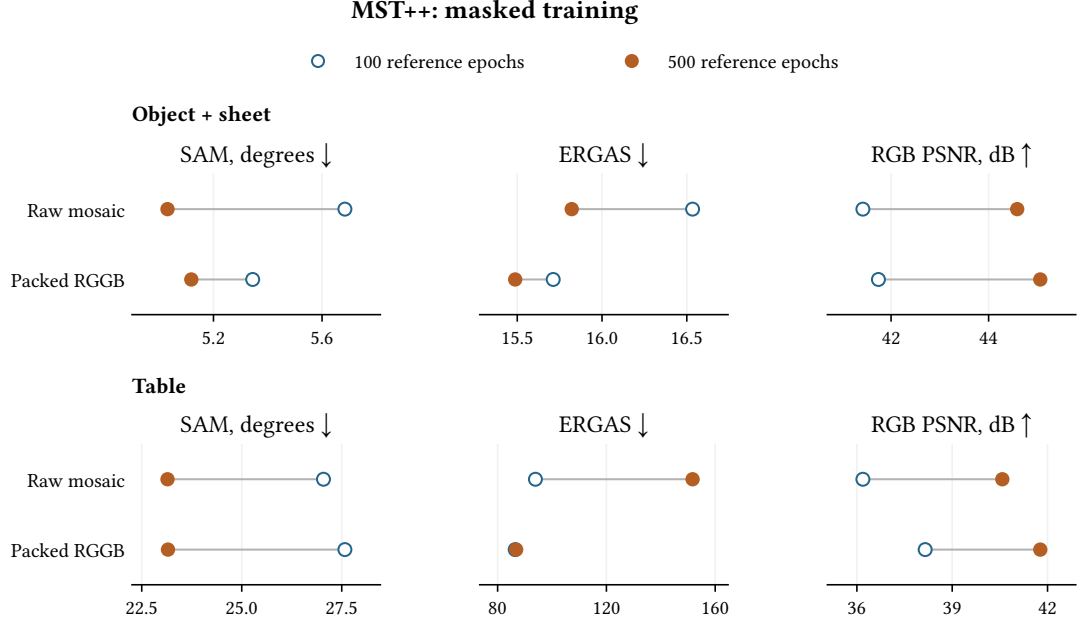


Figure 9: Masked raw-input and packed-input MST++ after 100 and 500 reference epochs. Each marker averages twelve validation images with predictions clipped to $[0, 1]$. SAM and ERGAS are lower-is-better; RGB PSNR is higher-is-better.

For raw-input MST++, mean SAM in the object + sheet region falls from 5.68 to 5.03 degrees, compared with 5.34 to 5.12 for packed input. At 500 epochs, packed input still has better mean ERGAS, PSNR, SSIM and color error, while SID is nearly equal. Raw input overtakes packed input in SSC and SAM.

For the table region, mean ERGAS rises from 93.91 to 151.69 with raw input, while packed-input ERGAS stays near 86.46–86.82. For every validation image in these four masked-model comparisons, ERGAS exceeds the local saturation threshold of about 41.45. Its transformed similarity is therefore 10^{-6} in every case. The ERGAS contribution to SSC cannot distinguish these substantially different errors, although the other score contributions can still change.

5.4 Ensembling

We form an ensemble by averaging MSFN and SPECAT predictions with equal weights, following SATE [11]. Both models use the common loss and the weights saved after 100 reference epochs. We pair models trained under the same policy, average their predictions, then clip the result to $[0, 1]$ for evaluation.

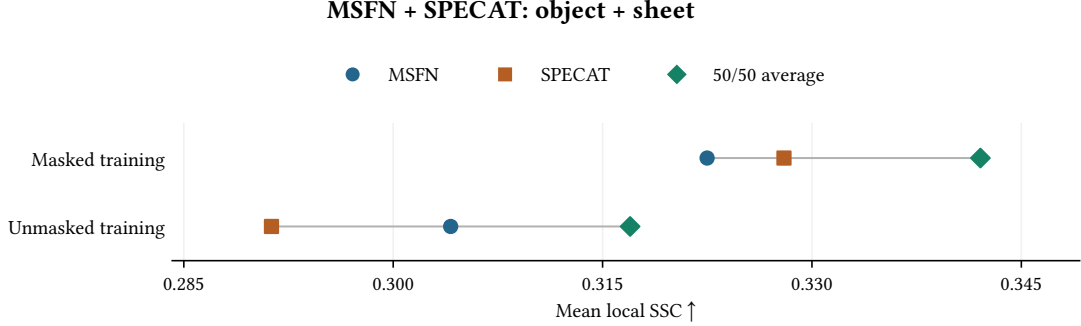


Figure 10: Object + sheet SSC for MSFN, SPECAT and their equally weighted prediction ensemble under each training policy. Every marker averages twelve images with predictions clipped to $[0, 1]$.

Under both training policies, the ensemble improves mean object + sheet SSC and all six mean submetrics relative to either model alone (Table 1). It has higher SSC than both individual models on nine of twelve validation images under masked training and ten of twelve under unmasked training.

Metric	Masked training			Unmasked training		
	MSFN	SPECAT	50/50 average	MSFN	SPECAT	50/50 average
SAM, degrees ↓	5.54	5.46	5.27	5.69	5.91	5.54
SID ↓	0.02273	0.02144	0.02006	0.02324	0.02570	0.02179
ERGAS ↓	15.93	15.32	15.22	16.96	16.97	16.44
RGB PSNR, dB ↑	40.51	40.62	41.52	40.32	39.25	40.55
RGB SSIM ↑	0.9791	0.9802	0.9829	0.9799	0.9771	0.9817
$\Delta E00$ ↓	0.946	0.907	0.825	0.902	1.052	0.880

Table 1: Object + sheet submetrics for MSFN, SPECAT and their equally weighted prediction ensembles after 100 reference epochs with the common loss. Every value averages the same twelve validation images with predictions clipped to $[0, 1]$. Arrows indicate the direction of improvement. Shading identifies the averages.

6 Discussion and conclusion

An SSC score of 0.5 or 0.1 needs an accompanying description of the reconstruction errors. Our spatial-averaging example gives full-frame SSC 0.502 for a precisely defined loss of detail (Appendix C.1). Two predictions with similar SSC can nevertheless have different spectral and spatial errors. The score alone cannot establish whether either prediction is suitable for a particular scientific measurement.

Regional evaluation reveals differences that full-frame reporting can obscure. The Gaussian-noise experiments show how a small absolute disturbance produces much larger spectral penalties in the dark table region than in object + sheet. In the training experiments, the policy that scores better changes with the evaluated region. Reporting the regions separately makes it possible to judge improvement in the part of the scene relevant to the task.

The individual metrics also change the interpretation of an SSC improvement. Masked raw-input MST++ has higher object + sheet SSC despite worse SAM, SID and ERGAS. Longer training changes table ERGAS substantially while its clipped contribution to SSC remains fixed. By contrast, the MSFN-SPECAT ensemble improves all six object + sheet metric means. The individual-image results show how consistently those gains occur across scenes.

Reproducible evaluation requires a versioned executable scorer, documented preprocessing and rendering, and explicit masks, clipping and aggregation rules. Reference inputs with expected metrics and scores would allow independent implementations to be checked. This would avoid much of the effort spent reconstructing the evaluation in this study. Appendix E records the discrepancies with the deployed scorer and the unresolved details.

Evaluation sets need enough examples to cover the intended scene and object variation. The released public set contains nine apple and two glass images; our Kaggle probes cover four released private inputs. We have not examined the separate hidden and additional final-assessment images, as they are kept private even after the challenge ended.

For this dataset, SSC should be reported alongside regional submetrics and per-image results. These show where a reconstruction improves, which errors account for the score change, and whether an average gain extends across scenes. Controlled changes to ground truth provide understandable examples of the errors that particular scores can represent.

References

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A Which images and reporting masks were used?

Figure 11 shows all twelve images selected for our validation set. The previews render the ground-truth cubes to RGB and apply a display-only brightening. Object names describe the visible contents. The secondary labels preserve the source identifiers: C2-0013, for example, denotes category 2, id 13. All twelve images come from the official training partition.

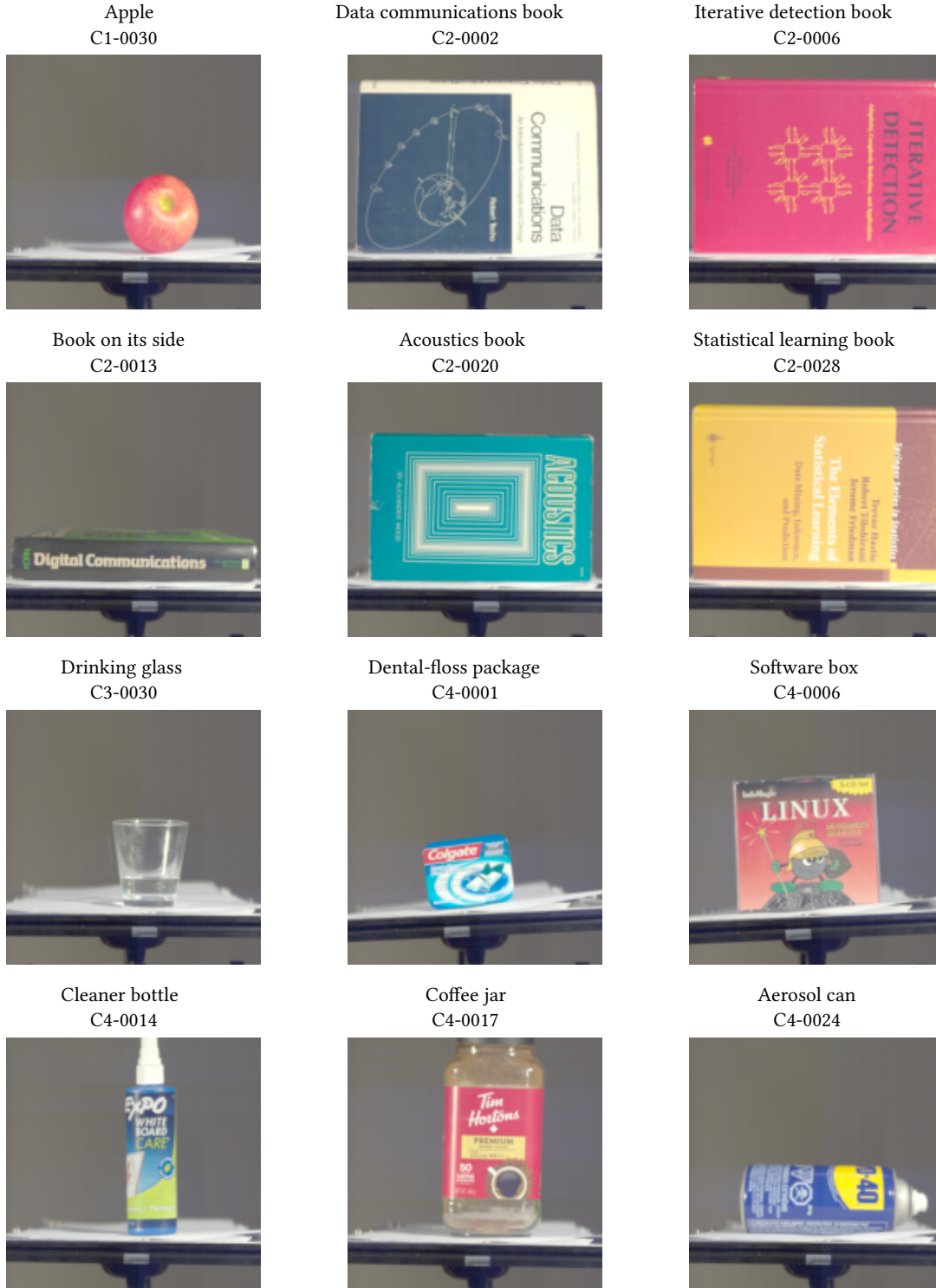


Figure 11: The twelve validation set images. The target previews retain the complete image fields, including the surrounding background. The previews use the same rendering and display brightening as the main-text examples.

The dataset identifiers group apples as Category 1, books as Category 2, drinking glasses as Category 3, and other household objects as Category 4. Our development allocation is one, five, one and five images respectively. The official public set contains nine apple and two glass images. The broader local development allocation was chosen to include book and household-object variation.

Of the 167 official training pairs, we use twelve for our validation set. One image is excluded because only an edge of the glass is in frame and there is no table region. The excluded image is marked X in Appendix A.2.3. The remaining 154 pairs and all eleven released public-test pairs form our 165-image training partition.

A.1 Mask construction

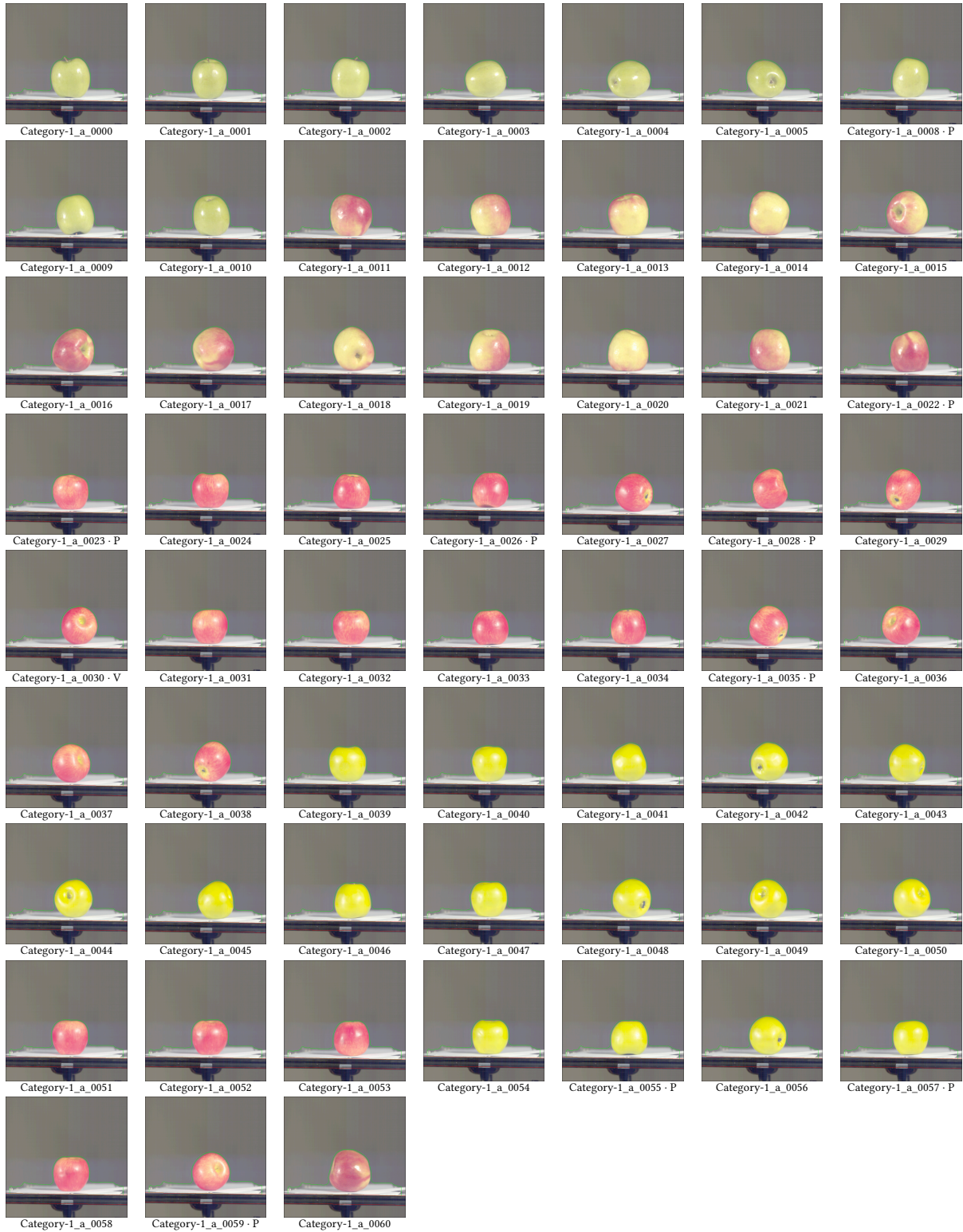
We construct the masks from target brightness, spatial morphology and a fitted sheet boundary, with convex hulls for books and six manual polygon overrides. Labels occupy a 512×512 grid; repeating each label over a two-by-two block aligns them with the full-resolution cubes.

A.2 Complete released-target atlas

The following sheets show all 178 released target images: 167 official training pairs and eleven public pairs. The four private inputs are shown in the main text; their targets and the final hidden targets are unavailable. Each target appears once here. Labels give its original partition and its membership in our local split.

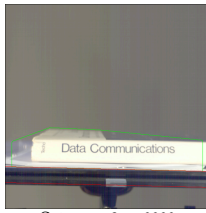
A.2.1 Apples: 59 images

50 official training; 9 public. Unmarked: official and local training · P: public, used for training · V: local validation



A.2.2 Books: 31 images

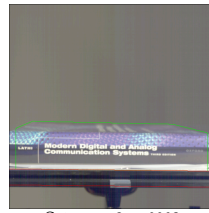
31 official training; 0 public. Unmarked: official and local training · P: public, used for training · V: local validation



Category-2_a_0000



Category-2_a_0002 · V



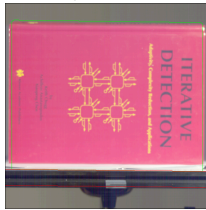
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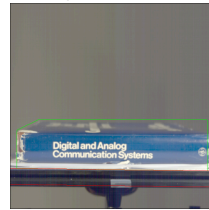
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Category-2_a_0005



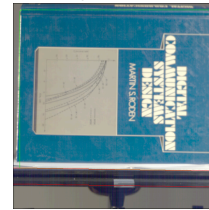
Category-2_a_0006 · V



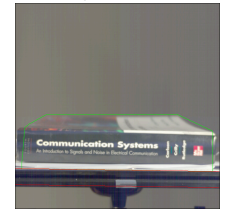
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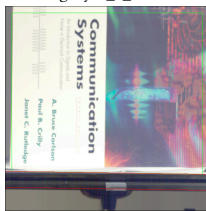
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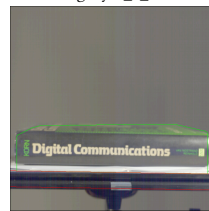
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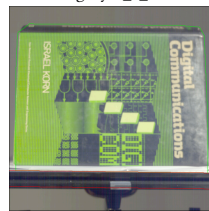
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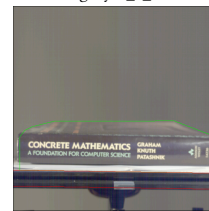
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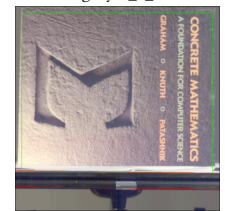
Category-2_a_0013 · V



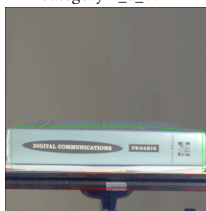
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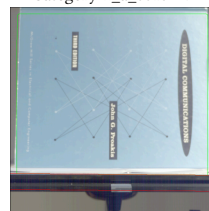
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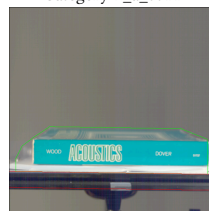
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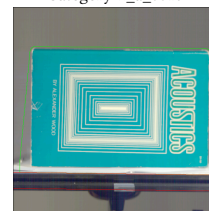
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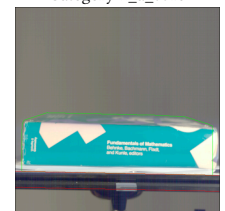
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Category-2_a_0019



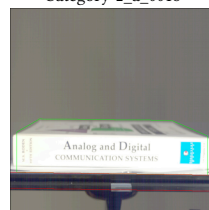
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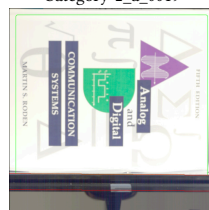
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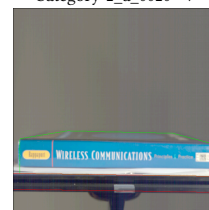
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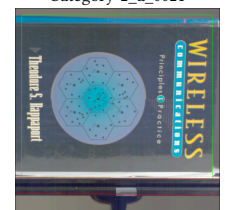
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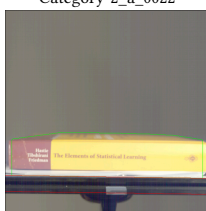
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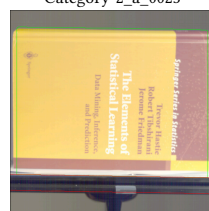
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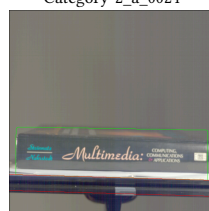
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Category-2_a_0027



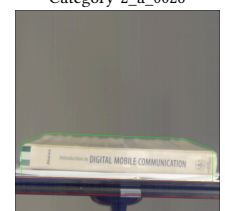
Category-2_a_0028 · V



Category-2_a_0029



Category-2_a_0030



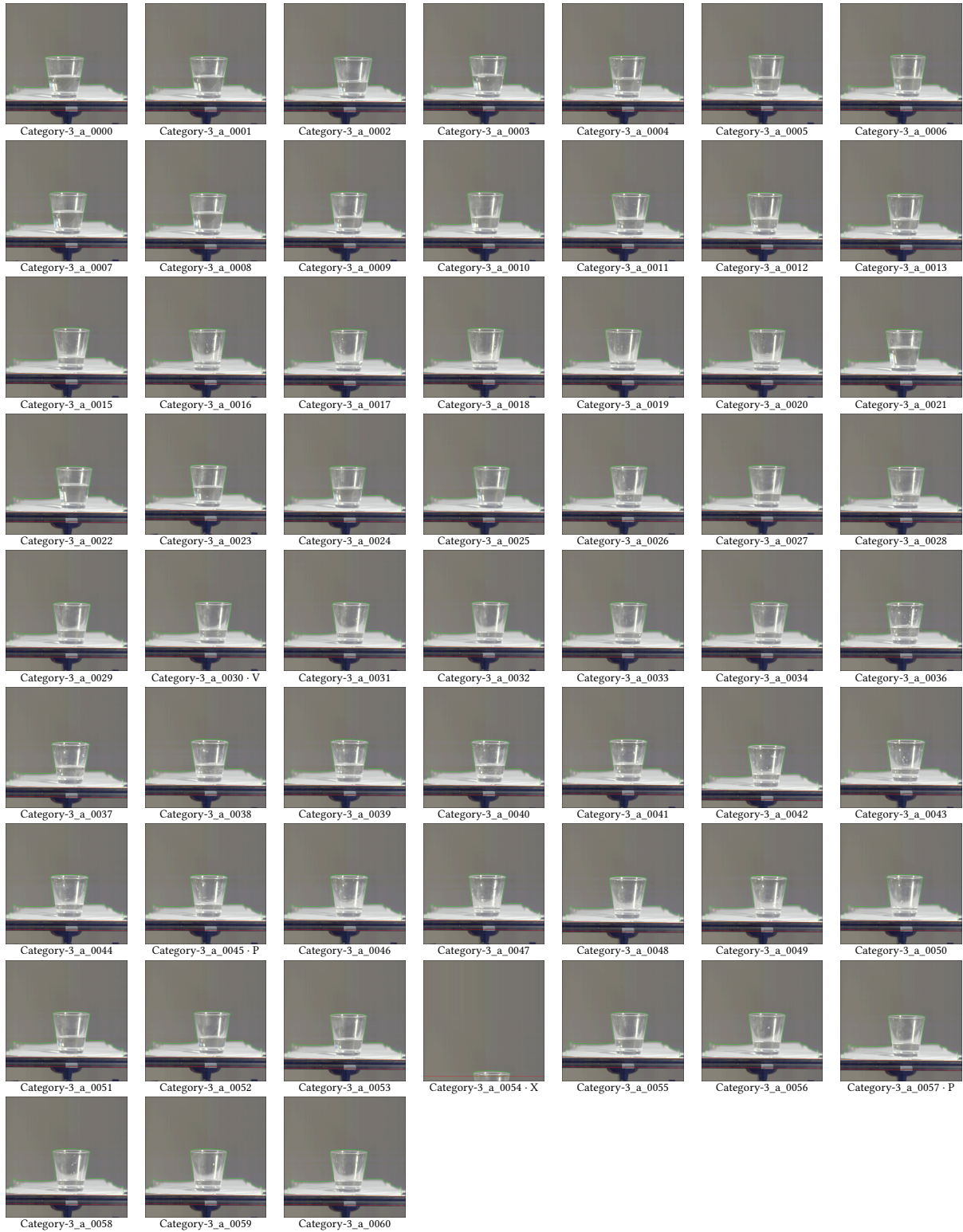
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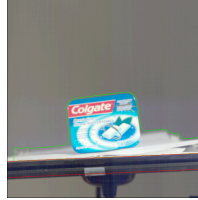
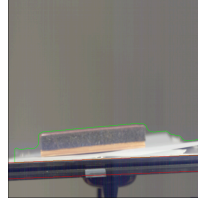
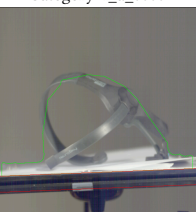
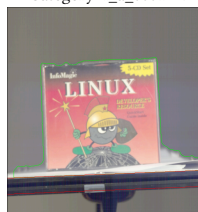
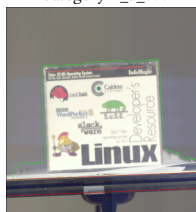
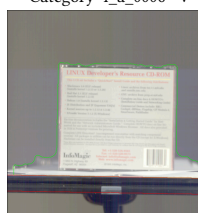
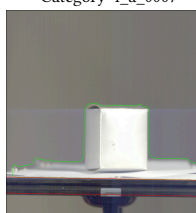
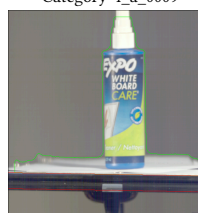
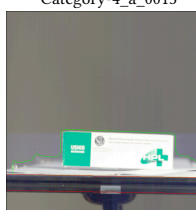
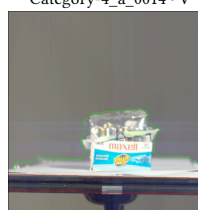
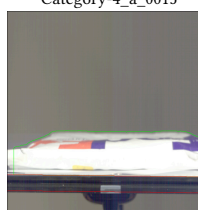
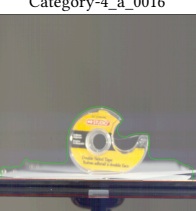
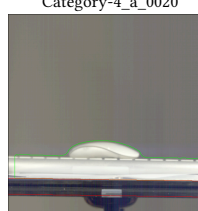
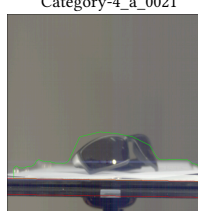
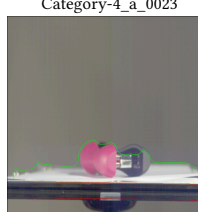
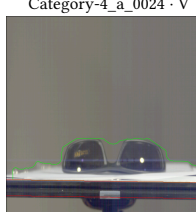
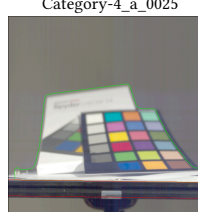
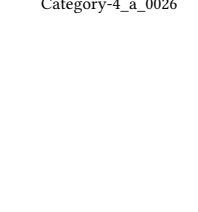
Category-2_a_0032

A.2.3 Drinking glasses: 59 images

57 official training; 2 public. Unmarked: official and local training · P: public, used for training · V: local validation · X: excluded



29 official training; 0 public. Unmarked: official and local training · P: public, used for training · V: local validation

				
Category-4_a_0000	Category-4_a_0001 · V	Category-4_a_0002	Category-4_a_0003	Category-4_a_0004
				
Category-4_a_0005	Category-4_a_0006 · V	Category-4_a_0007	Category-4_a_0009	Category-4_a_0010
				
Category-4_a_0011	Category-4_a_0012	Category-4_a_0013	Category-4_a_0014 · V	Category-4_a_0015
				
Category-4_a_0016	Category-4_a_0017 · V	Category-4_a_0019	Category-4_a_0020	Category-4_a_0021
				
Category-4_a_0022	Category-4_a_0023	Category-4_a_0024 · V	Category-4_a_0025	Category-4_a_0026
				
Category-4_a_0027	Category-4_a_0028	Category-4_a_0029	Category-4_a_0030	

B How are the local metrics computed?

Let g_{ib} and p_{ib} be target and evaluated prediction values at pixel i and band b , with $B = 61$. Let R be the selected region and $n = |R|$. The following formulas specify the spectral metrics in the local floating-point implementation.

The spectral angle at a pixel is

$$a_i = \frac{180}{\pi} \arccos \left(\min \left(1, \max \left(-1, \frac{\sum_{b=1}^B g_{ib} p_{ib}}{\sqrt{\sum_{b=1}^B g_{ib}^2} \sqrt{\sum_{b=1}^B p_{ib}^2 + 10^{-8}}} \right) \right) \right), \quad (7)$$

$$\text{SAM} = \frac{1}{n} \sum_{i \in R} a_i.$$

The denominator epsilon assigns a 90-degree angle to an all-zero prediction. It also produces nonzero angles for identical nonzero spectra, which can be large when the spectra are sufficiently dark. For example, two identical 61-band spectra with every value equal to 10^{-5} give an angle of approximately 67.7° . This algebraic example illustrates the formula; its prevalence in the data was not measured. The training loss retains the denominator epsilon and additionally adds 10^{-8} to every spectral value in both inputs. It clips the cosine to $[-1 + 10^{-7}, 1 - 10^{-7}]$ and retains angles in radians.

For SID, with $\varepsilon = 10^{-12}$, define

$$q_{ib} = \frac{g_{ib}}{\sum_{c=1}^B g_{ic} + \varepsilon}, \quad r_{ib} = \frac{p_{ib}}{\sum_{c=1}^B p_{ic} + \varepsilon}. \quad (8)$$

Then

$$\text{SID} = \frac{1}{n} \sum_{i \in R} \sum_{b=1}^B \left[q_{ib} \ln \left(\frac{q_{ib} + \varepsilon}{r_{ib} + \varepsilon} \right) + r_{ib} \ln \left(\frac{r_{ib} + \varepsilon}{q_{ib} + \varepsilon} \right) \right]. \quad (9)$$

The epsilon makes zero-band penalties finite; it does not remove their dependence on the other band values.

For ERGAS, let $\mu_b = \frac{1}{n} \sum_{i \in R} g_{ib}$ and $e_b^2 = \frac{1}{n} \sum_{i \in R} (p_{ib} - g_{ib})^2$. The evaluator uses scale factor one:

$$\text{ERGAS} = 100 \sqrt{\frac{1}{B} \sum_{b=1}^B \frac{e_b^2}{(|\mu_b| + 10^{-12})^2}}. \quad (10)$$

The renderer converts the 400 to 700 nm bands to sRGB using D65 illumination and the CIE 1931 two-degree observer. It clips negative linear RGB values to zero, applies the sRGB transfer function and clips the result to $[0, 1]$. These steps belong to numerical evaluation, before any brightening of figures.

For RGB images G and P , the regional error is $\text{MSE} = \frac{1}{3n} \sum_{i \in R} \sum_{c=1}^3 (P_{ic} - G_{ic})^2$, and $\text{PSNR} = -10 \log_{10}(\max(\text{MSE}, 10^{-12}))$. The numerical lower bound caps PSNR at 120 dB. SSIM uses an 11×11 Gaussian window with standard deviation 1.5 pixels and reflected image boundaries. Its constants are $C_1 = 0.01^2$ and $C_2 = 0.03^2$, with 10^{-12} added to its denominator; window moments use Gaussian weights without a sample-covariance correction. Channel averaging precedes regional averaging. CIEDE2000 uses the standard unit weighting factors and a D65 reference white.

The spatial SSC group can be negative because the implementation does not clip SSIM to zero. In that case its fractional power does not define a real SSC.

B.1 Regions, prediction postprocessing and aggregation

ERGAS recomputes target means within each selected region. SSIM is calculated on the full image before map positions inside the region are selected, so its 11×11 windows can include neighboring pixels outside the mask.

For trained predictions, we evaluate clipping to $[0, 1]$ and flooring at 0.005 with the same upper cap one. The target remains unchanged. Perturbation operations retain the separate construction rules in Appendix C.

Each image’s regional metrics and SSC are calculated before taking equal-image means. Region size does not change an image’s weight. Applying SSC to averaged raw metrics would generally give a different result. Regional SSCs are not additive parts of full-frame SSC.

C What do the complete perturbation results show?

Our study evaluates 43 settings on twelve validation cubes, producing 2064 image/setting/region records. All operations act on the full $1024 \times 1024 \times 61$ target cube. The altered cube is clipped below zero before scoring; there is no general upper clip. The scorer and aggregation are the same as those used in the main text. The following family labels correspond to subsections C.1–C.9.

Family	Operation	Settings and construction
M1	Pixel replacement and pooling	Random replacement probabilities 0.01, 0.1, 0.5. Each selected spectrum becomes the mean of four adjacent spectra from the original cube, with reflected boundaries. Separately, nonoverlapping 2×2 , 4×4 and 8×8 block averages are bilinearly resized to the original size with aligned corners disabled.
M2	Band interpolation	Dropped zero-based indices start at $\frac{e}{2}$ and advance by e , for $e = 10, 4, 2$. Each value is interpolated linearly between nearest retained bands. The three settings remove 6, 15 and 30 bands respectively.
M3	PCA projection	Keep 30, 12, 8, 4 or 2 components from the training-derived basis. Each validation spectrum is centered by the training mean, projected and reconstructed before clipping.
M4	Gaussian noise	Independent bandwise additive draws at $\sigma=0.001, 0.003, 0.01, 0.03$ and 0.05 reflectance units. Negative clipping is applied after addition.
M5	Spatial Gaussian smoothing	$\sigma=0.5, 1, 2$ and 4 pixels, applied separately to every band.
M6	Column-profile subtraction	Subtract a high-pass column profile derived either from the same image or from the training mean column profile. The low-pass profile uses Gaussian $\sigma=8$ columns.
M7	Gain	Global $\pm 1\%, \pm 3\%, \pm 10\%$; per-band $\pm 1\%, \pm 3\%, \pm 10\%$ with one fixed random sign per band. Positive outputs remain uncapped.
M8	Thresholds and floors	Set individual values below 0.001, 0.003, 0.01 or 0.02 to zero. Separately, raise values to floors 0.005 and 0.02.
M9	Constant cubes	Every pixel receives zero, the sampled training mean spectrum, or one.

The PCA coefficients for validation are obtained from the validation target itself. Thus the projection tests how the representation preserves target spectra. Its explained variance refers to the sampled training spectra. Four components retain 99.27% of that variance.

The random-number generators use seed 0. The study records one realization for each random setting/image, without an uncertainty estimate over repeated draws. Per-band gain signs remain fixed across images. The brightness analysis uses a separate Gaussian draw at $\sigma=0.01$.

Complete regional means

The charts in C.1–C.9 show all six raw metrics and SSC for every setting and region.

C.1 M1: Pixel replacement and pooling

Regional means: Figure 14 and Figure 15.

C.1.1 Pixel replacement, pooling and smoothing

Random pixel replacement changes selected spectra to the mean of their four immediate neighbors in the original cube. We use replacement probabilities 0.01, 0.1 and 0.5. Pooling instead averages every nonoverlapping 2×2 , 4×4 or 8×8 block, then bilinearly resizes the result to 1024×1024 . Every output pixel changes. The gap between these groups in Figure 12 separates two different operations.

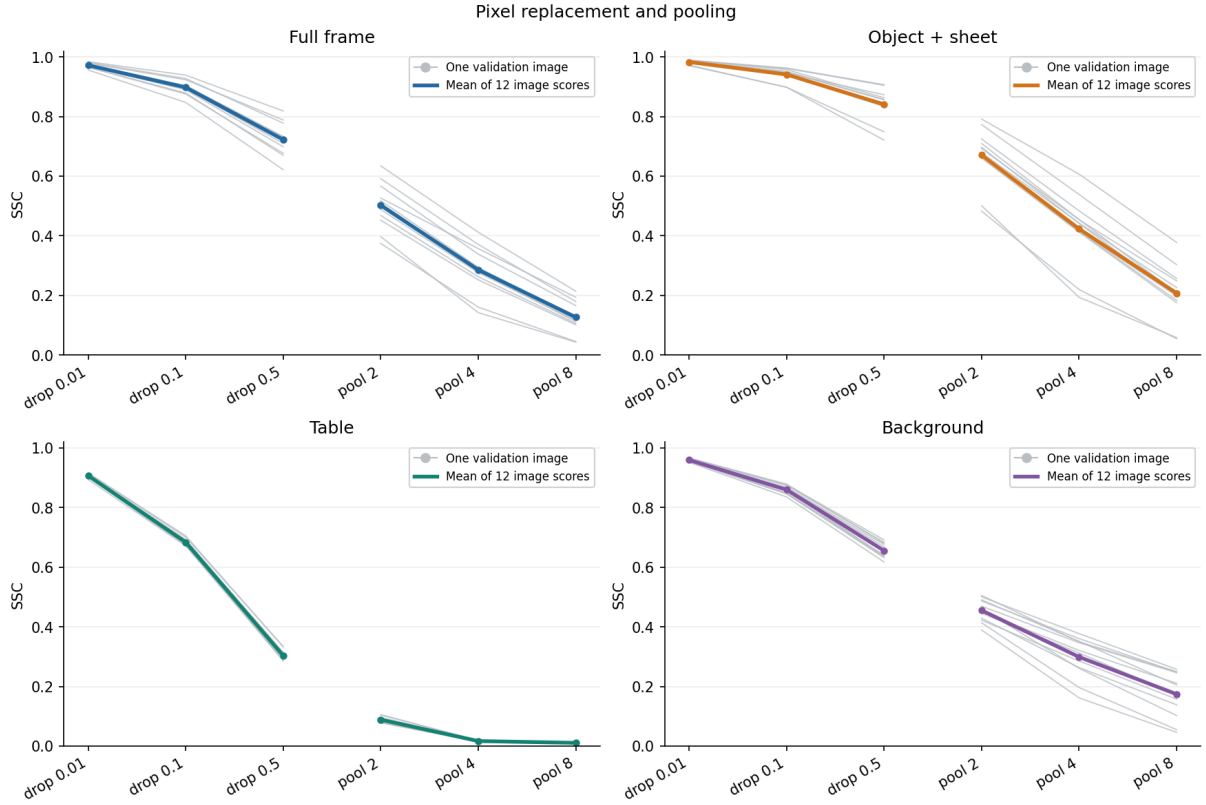


Figure 12: Pixel replacement and pooling in all four regions. Gray curves show individual images; colored curves show equal-image means. Labels “drop” give the random replacement probability and “pool” the block side length.

Replacing one pixel in a hundred leaves mean object + sheet SSC at 0.983. Pooling by two reduces it to 0.671, with individual scenes ranging from 0.482 to 0.791. The lowest and highest values belong to the book lying on its side and the apple, respectively. Their full-frame scores are 0.374 and 0.526.

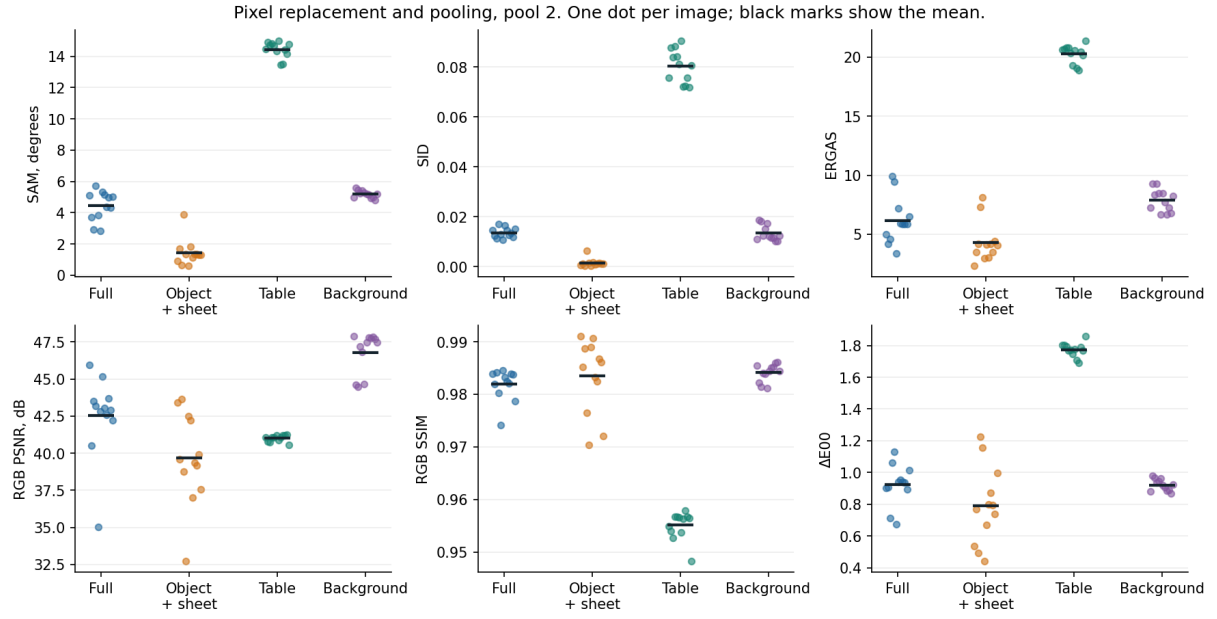


Figure 13: The six metrics after 2×2 pooling and bilinear resizing. Each dot represents an image; black marks show the means. Lower is better for SAM, SID, ERGAS and ΔE_{00} , and higher for RGB PSNR and SSIM.

At 2×2 pooling, object + sheet SAM is 1.44° , ERGAS is 4.31 and RGB PSNR is 39.7 dB. Table SAM rises to 14.43° and ERGAS to 20.26, despite RGB SSIM of 0.955 there. Full-frame RGB SSIM remains 0.982.

M1 · Random pixel replacement

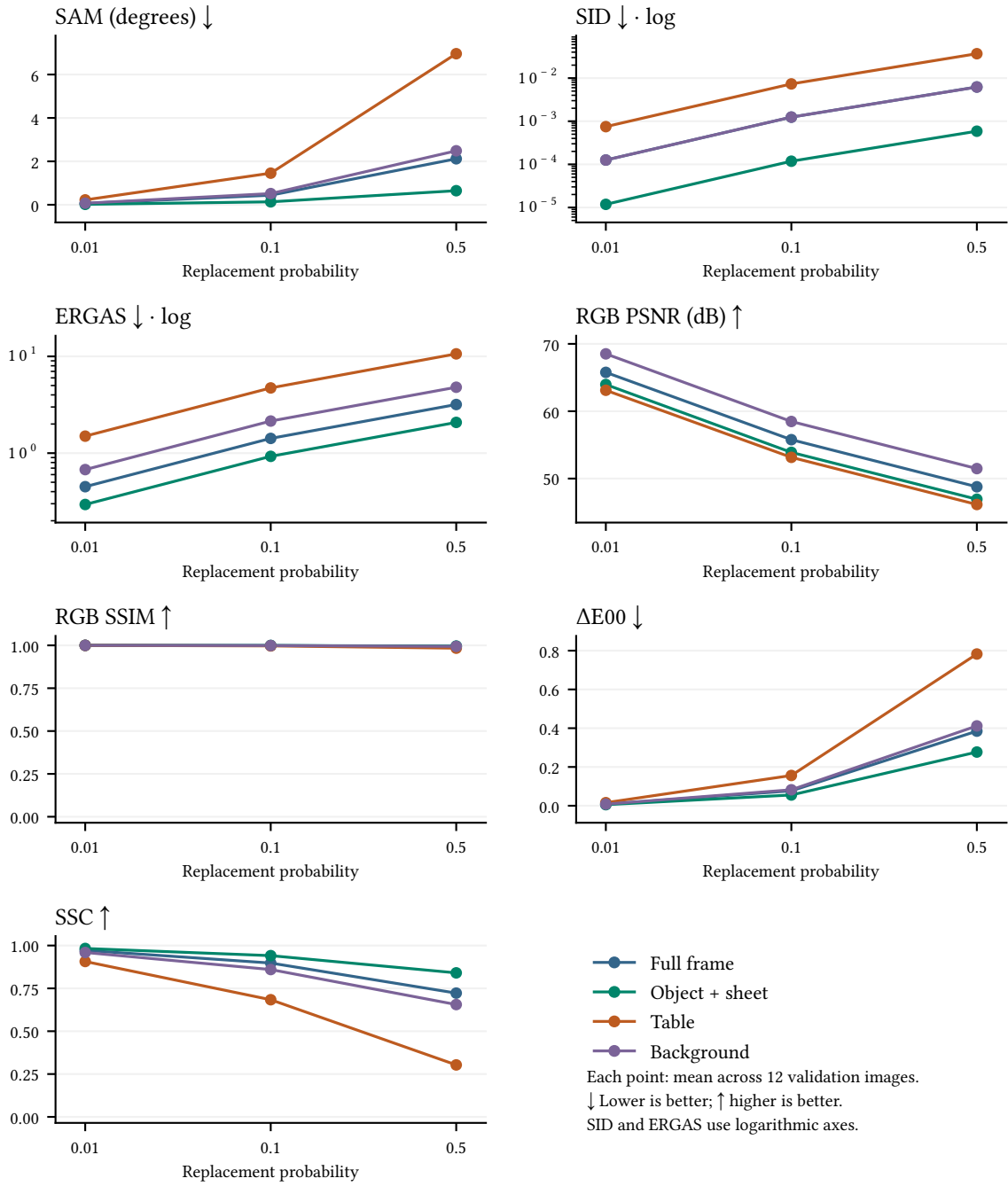


Figure 14: M1: Random pixel replacement. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

M1 · Spatial pooling and resizing

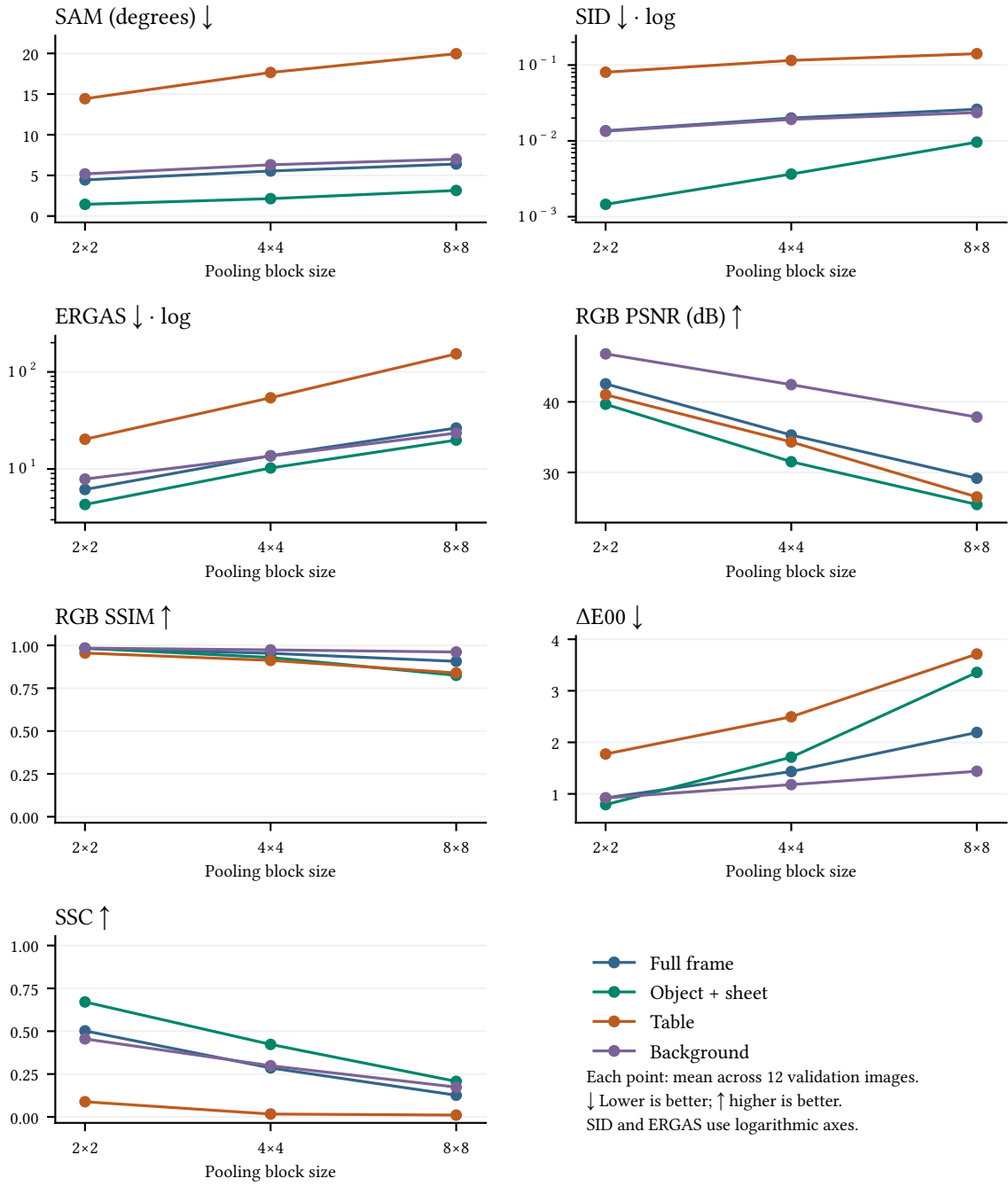


Figure 15: M1: Spatial pooling and resizing. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.2 M2: Band interpolation

Regional means: Figure 16.

C.2.1 Further spectral-detail comparisons

The six-band interpolation setting replaces 450, 550, 650, 750, 850 and 950 nm. The fifteen-band setting replaces 420, 460, 500, 540, 580, 620, 660, 700, 740, 780, 820, 860, 900, 940 and 980 nm. Their mean full-frame SSCs are 0.498 and 0.603. Figure 16 shows all six raw metrics and SSC for these band-interpolation settings.

M2 · Band interpolation

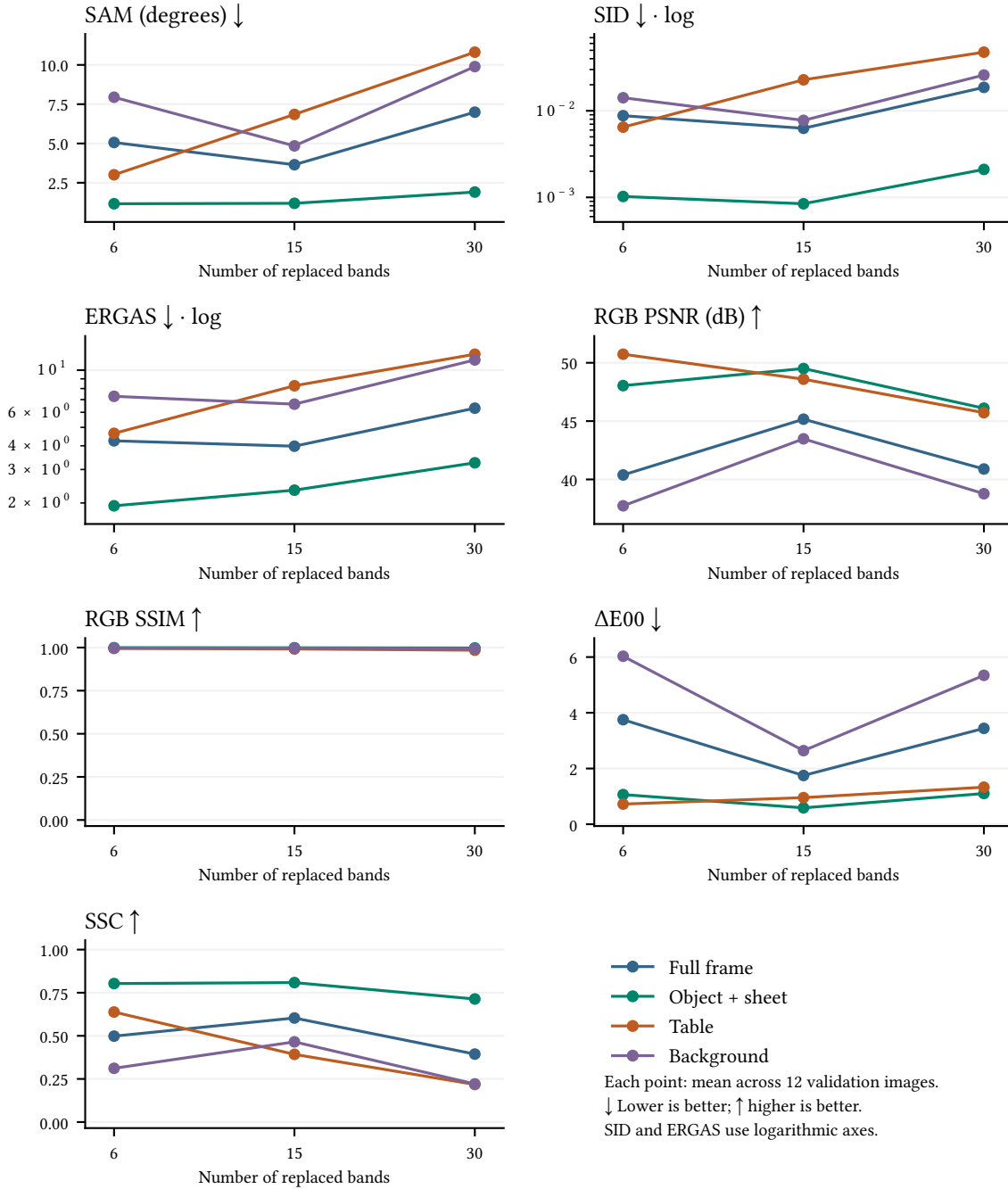


Figure 16: M2: Band interpolation. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.3 M3: PCA projection

Regional means: Figure 17.

Four PCA components retain 99.27% of sampled training variance, but the clipped validation projection has mean SSC 0.274 on object + sheet and 0.024 over the full frame. The coefficients come from each validation target.

M3 · PCA projection

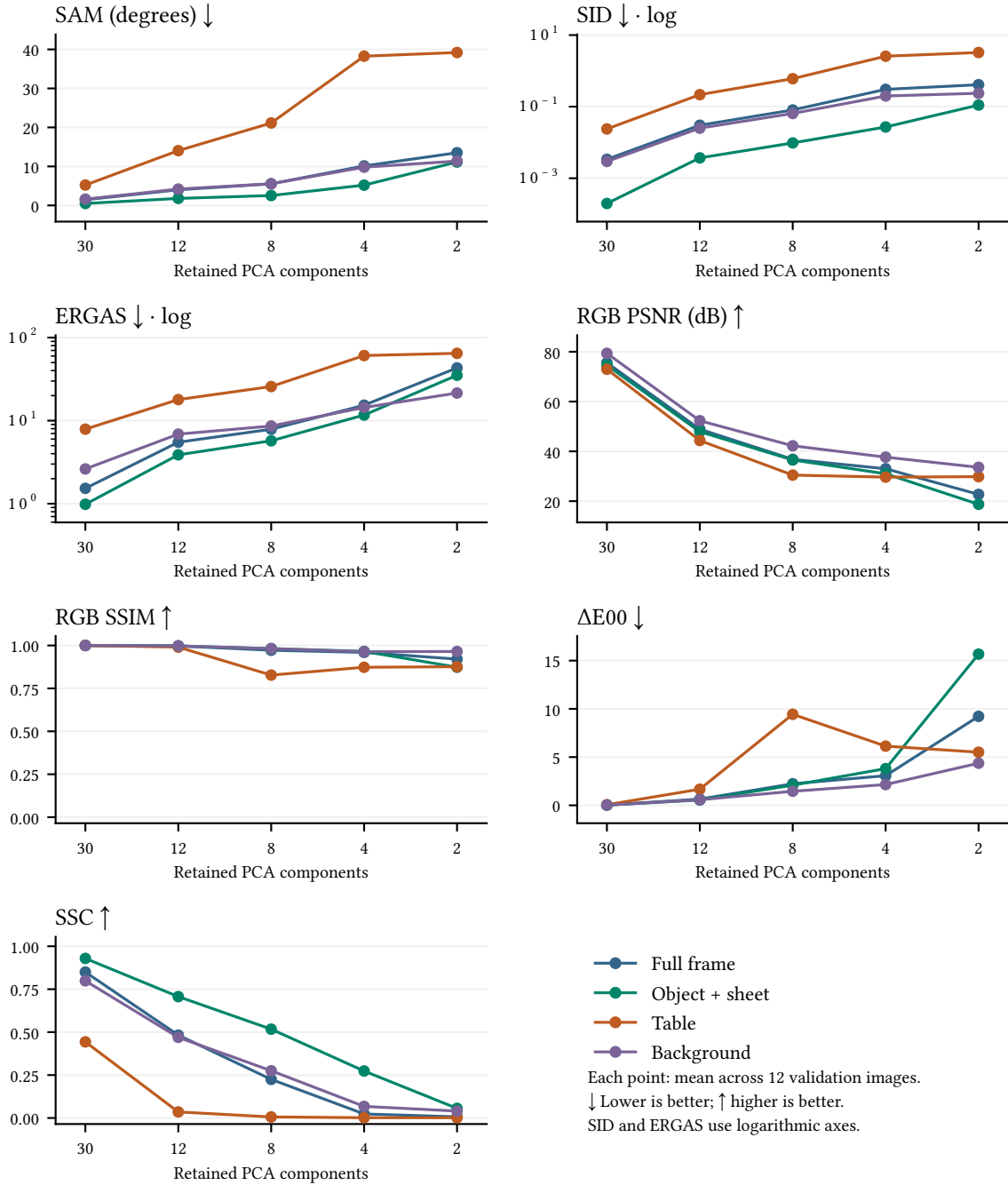


Figure 17: M3: PCA projection. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.4 M4: Gaussian noise

Regional means: Figure 19.

C.4.1 Sensitivity within brightness groups

A separate recorded draw at $\sigma=0.01$ lets us examine brightness within each image. Pixels are ordered by mean target reflectance and divided into ten estimated brightness groups. Mean SAM is about 31.9° in the darkest group and 1.0° in the brightest; mean SID falls from about 3.83 to 0.0008. Figure 18 retains each scene's curve. The groups are ranks within an image, so the same group number can cover different reflectance intervals across scenes.

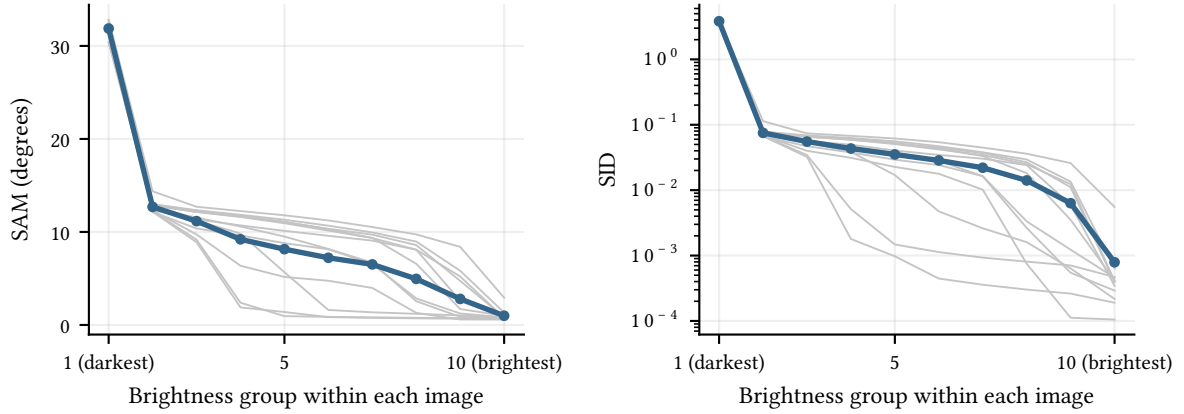


Figure 18: Spectral error by brightness rank under $\sigma=0.01$ noise. Each gray curve represents one image; blue gives the equal-image mean. SID uses a logarithmic vertical axis. The lowest-brightness groups incur much larger spectral penalties under this absolute disturbance.

This comparison orders pixels within each image rather than pooling bright and dark scenes into separate groups.

Region	SAM °	SID	ERGAS	PSNR dB	SSIM	ΔE_{00}
Full frame	9.56	0.4111	7.49	38.39	0.9074	2.47
Object + sheet	2.53	0.0094	3.30	43.04	0.9647	1.07
Table	30.26	3.5452	56.54	31.95	0.7138	5.85
Background	11.59	0.2750	18.28	38.62	0.8986	2.96

Table 2: Regional means from the main $\sigma=0.01$ perturbation sweep, after clipping negative values. These use the sweep realization, separate from the brightness-group draw above.

M4 · Gaussian noise

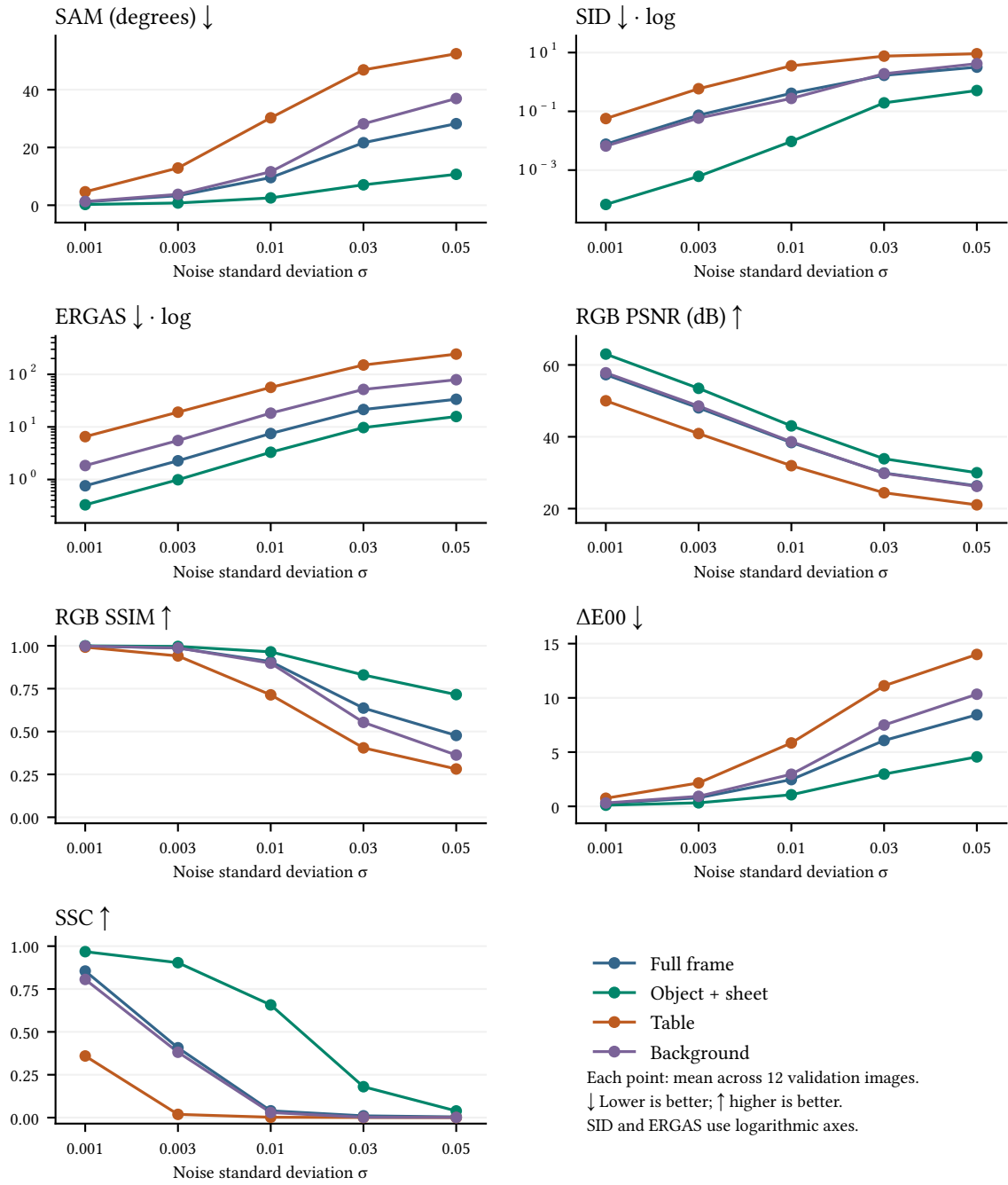


Figure 19: M4: Gaussian noise. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.5 M5: Spatial Gaussian smoothing

Regional means: Figure 20.

M5 · Spatial Gaussian smoothing

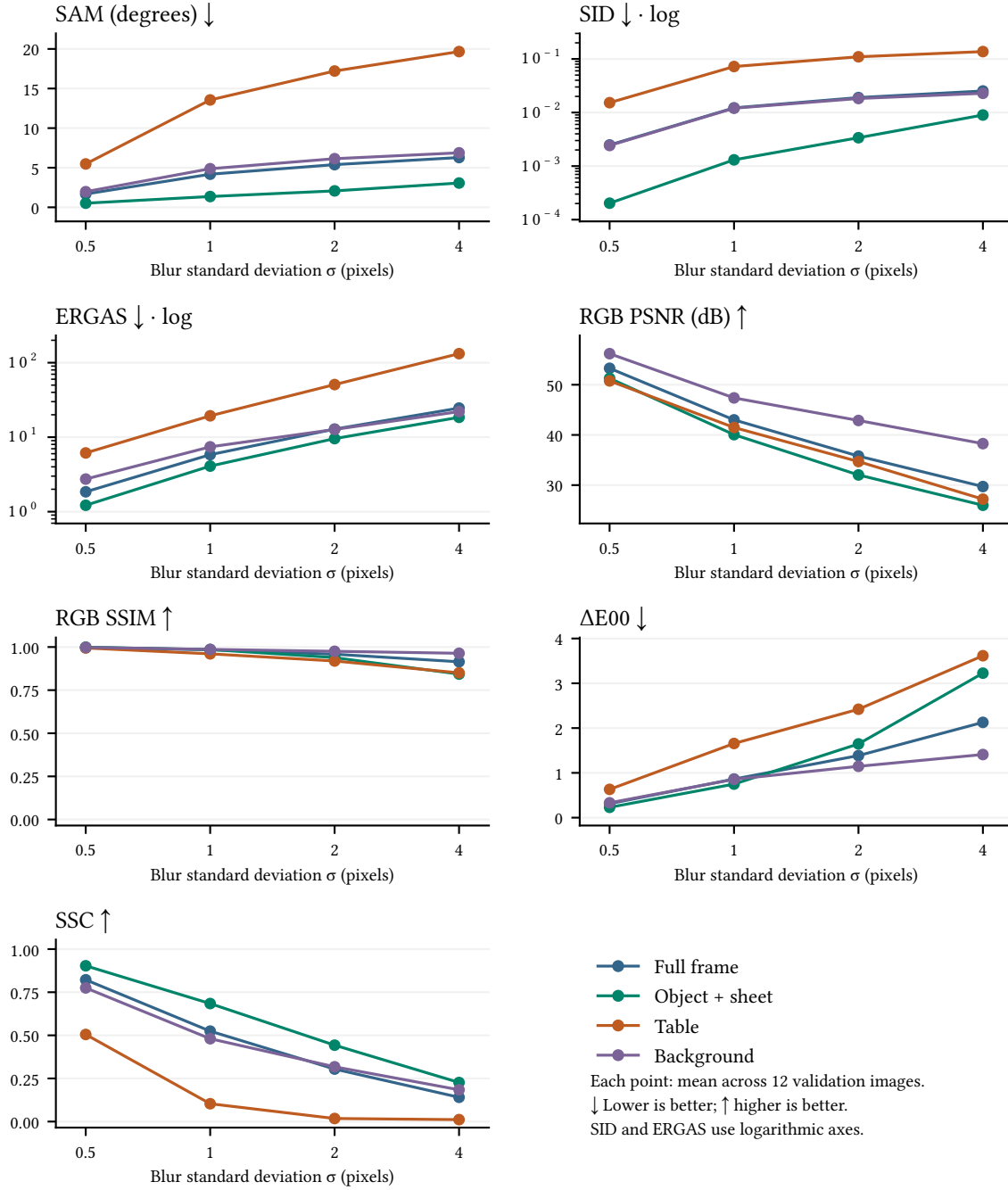


Figure 20: M5: Spatial Gaussian smoothing. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.6 M6: Column-profile subtraction

Regional means: Figure 21.

M6 · Column-profile subtraction

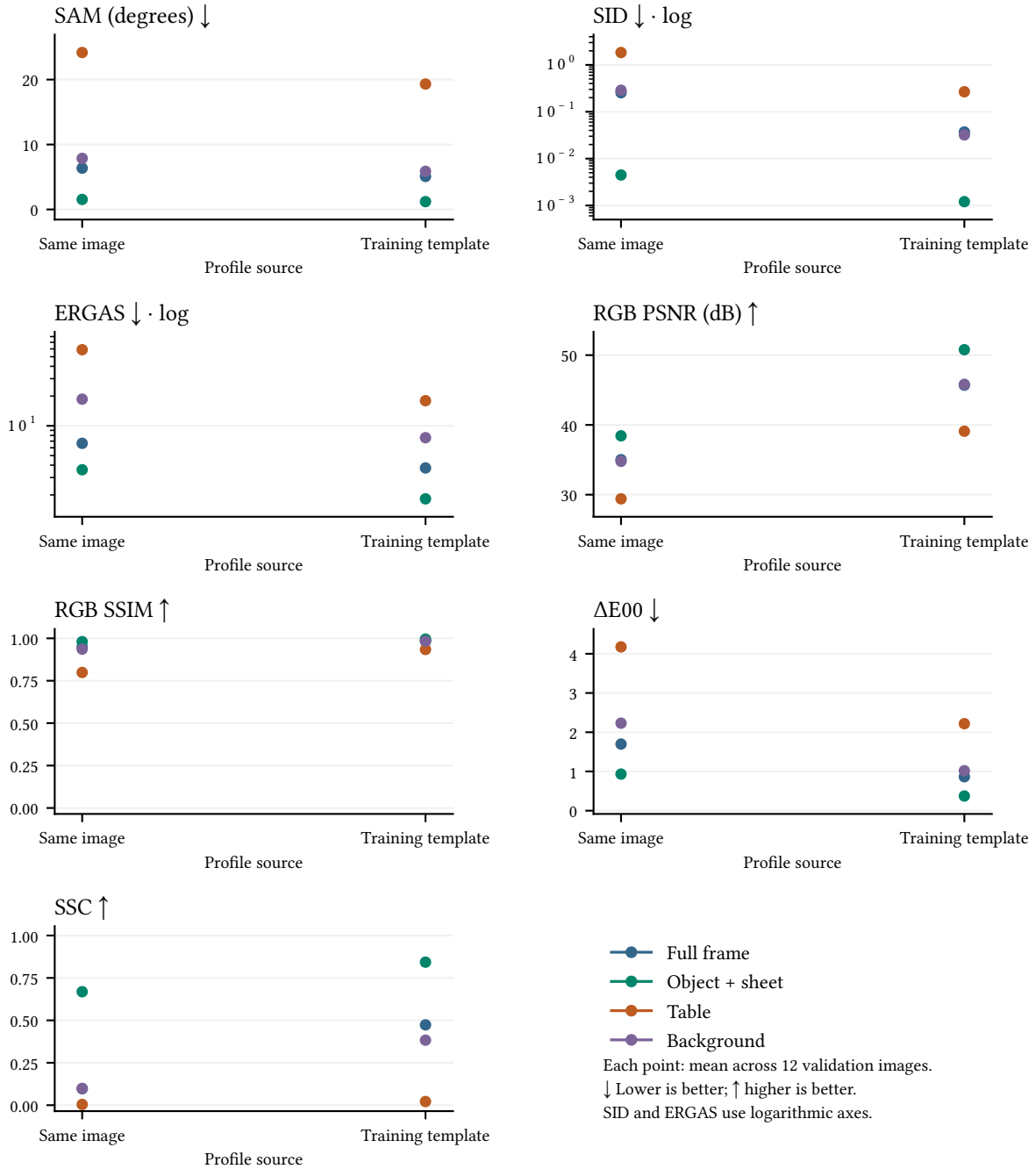


Figure 21: M6: Column-profile subtraction. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images.

C.7 M7: Gain

Regional means: Figure 22 and Figure 23.

C.7.1 Gain separates scale from spectral shape

A global gain multiplies every value by the same positive factor. In the ideal spectral-angle and normalized-divergence definitions, this preserves spectral direction and band proportions. The implemented stabilizers leave small numerical residuals. Amplitude-sensitive metrics still respond.

At a gain of +3%, object + sheet SAM is about 0.01° and SID rounds to zero at four decimals. ERGAS is 3.75, RGB PSNR is 42.6 dB and ΔE_{00} is 0.61. Mean SSC is 0.751 on object + sheet and 0.743 on table. This close regional response differs markedly from the additive-noise result, because the absolute gain error scales with the original signal. Object + sheet SSC spans only 0.725 to 0.761 across scenes at this setting.

Per-band gains change band proportions too. With one fixed pattern of +3% and -3% gains across bands, mean object + sheet SAM rises to 1.69° and SSC falls to 0.677. This comparison explains why a good spectral-angle result can coexist with amplitude error, and why checking several metrics is useful. The experiment uses one fixed sign pattern across all twelve images.

M7 · Global gain

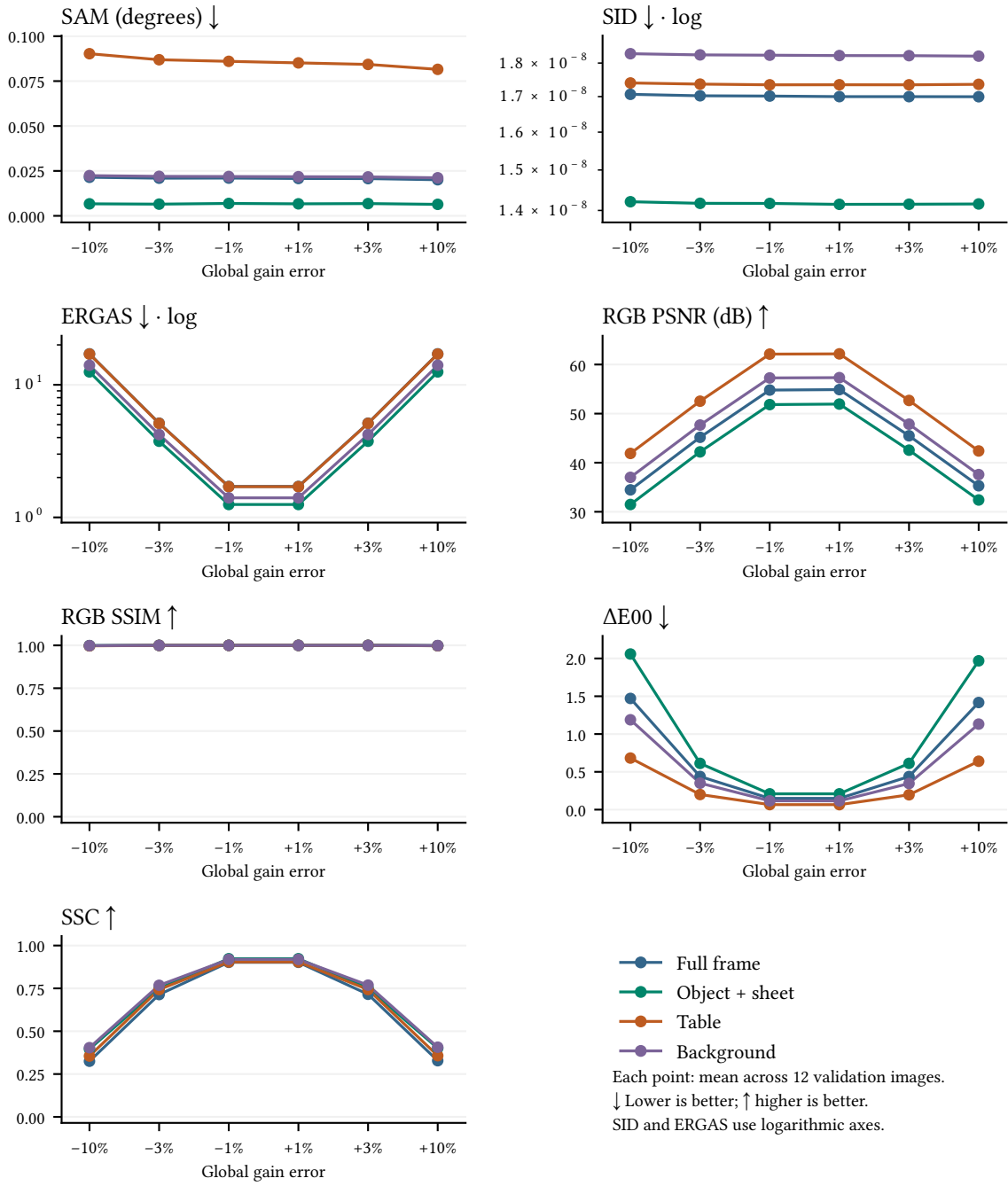


Figure 22: M7: Global gain. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

M7 · Per-band gain

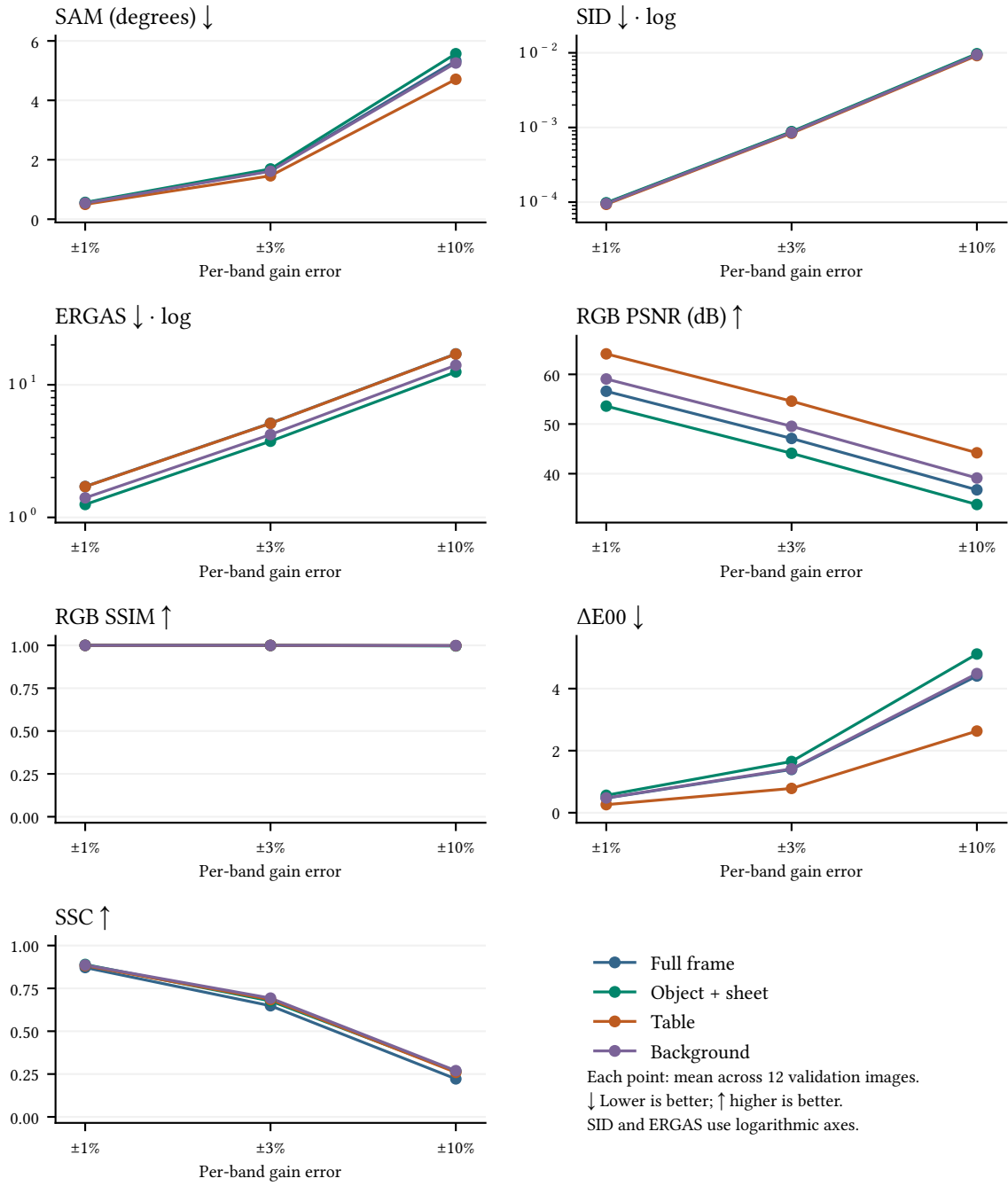


Figure 23: M7: Per-band gain. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.8 M8: Thresholds and floors

Regional means: Figure 24 and Figure 25.

C.8.1 Zeros and positive floors

Thresholding replaces small band values with exact zeros; flooring raises them to a positive minimum. Zeroing values below 0.02 gives mean object + sheet SSC 0.889, but the book-on-its-side scene scores 0.099 while the apple scores 0.998. In the book region, 15.24% of pixels contain at least one target band below the threshold, compared with 0.113% for the apple.

Flooring the target at 0.005 leaves mean object + sheet SSC close to one but reduces table SSC to 0.388. Applying the same floor to an erroneous model prediction can have a different effect, particularly when a bright spectrum contains bands predicted as zero. Appendix D.3 examines that case in trained predictions.

C.8.2 Selected individual-image references

The following table preserves three reference operations by scene. These are complete twelve-image comparisons, with full-frame and object + sheet SSC shown separately.

Scene	Noise full	Noise object	Pool full	Pool object	Zero full	Zero object
C1-0030	0.400	0.932	0.526	0.791	0.062	0.998
C2-0002	0.418	0.899	0.566	0.693	0.069	0.906
C2-0006	0.433	0.887	0.592	0.725	0.066	0.988
C2-0013	0.319	0.779	0.374	0.482	0.058	0.099
C2-0020	0.426	0.915	0.397	0.501	0.068	0.958
C2-0028	0.473	0.937	0.634	0.773	0.070	0.997
C3-0030	0.330	0.892	0.452	0.678	0.062	0.984
C4-0001	0.397	0.938	0.488	0.710	0.065	0.969
C4-0006	0.432	0.914	0.510	0.669	0.067	0.997
C4-0014	0.415	0.919	0.500	0.697	0.064	0.861
C4-0017	0.452	0.909	0.517	0.669	0.065	0.988
C4-0024	0.393	0.921	0.468	0.660	0.063	0.925

Table 3: Individual-image SSC anchors. Noise uses $\sigma=0.003$, pooling uses 2×2 blocks followed by bilinear resizing, and zeroing removes values below 0.02. Object includes the sheet.

For the threshold example above, affected band values account for 1.49% in the book-on-its-side region and 0.0021% in the apple region.

M8 · Zero thresholds

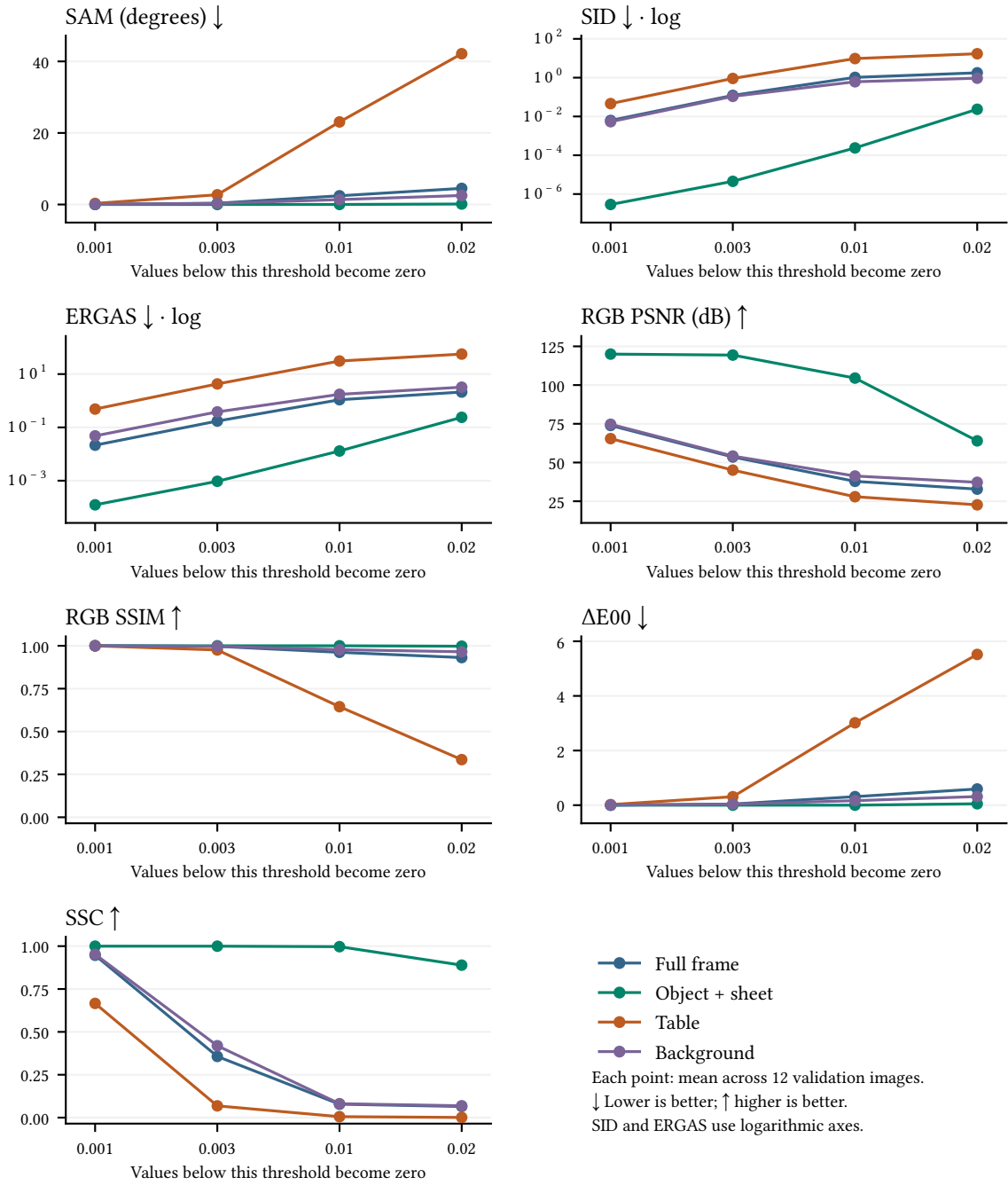


Figure 24: M8: Zero thresholds. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

M8 · Positive floors

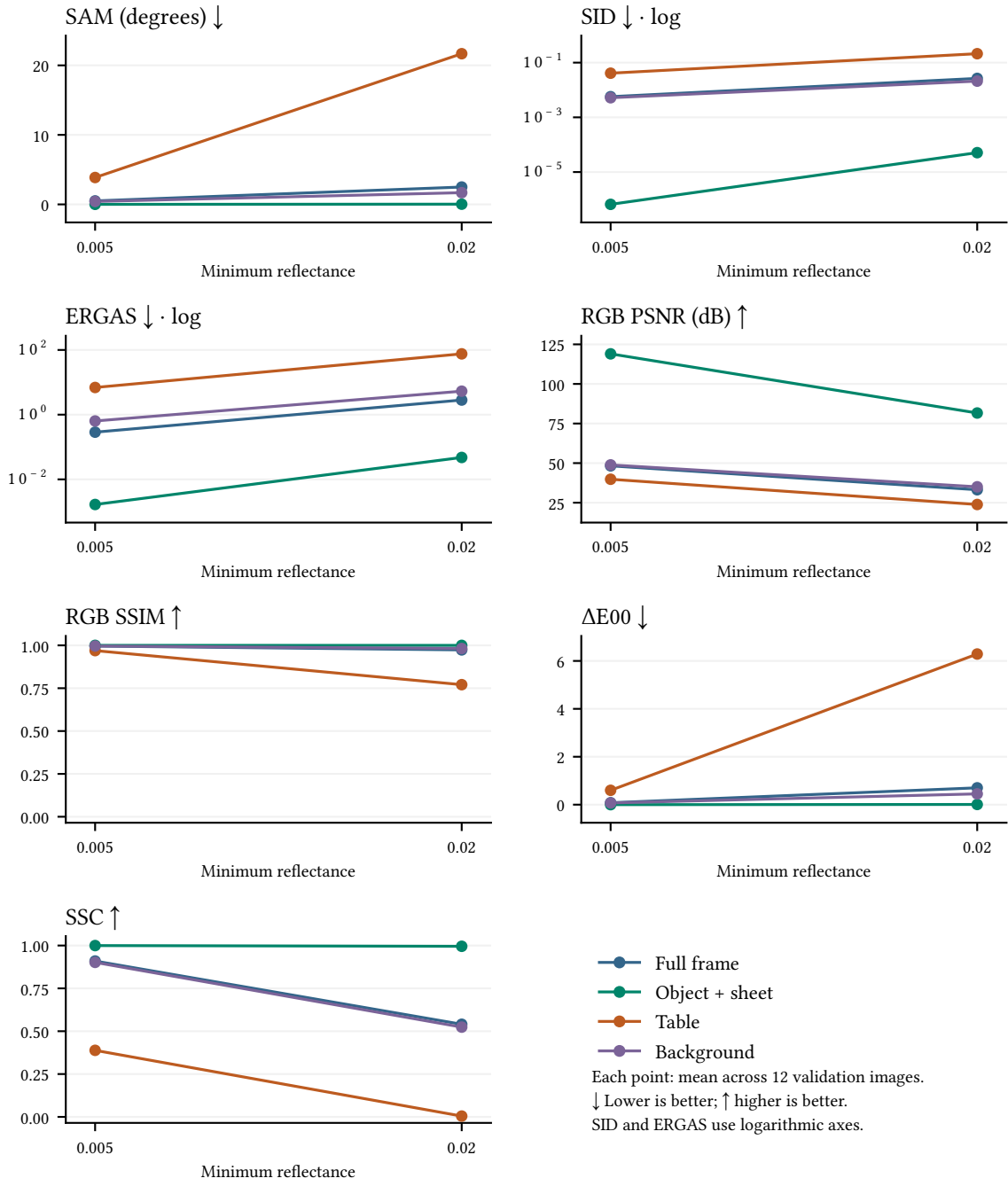


Figure 25: M8: Positive floors. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images. Settings are spaced equally along the horizontal axis.

C.9 M9: Constant cubes

Regional means: Figure 26.

M9 · Constant cubes

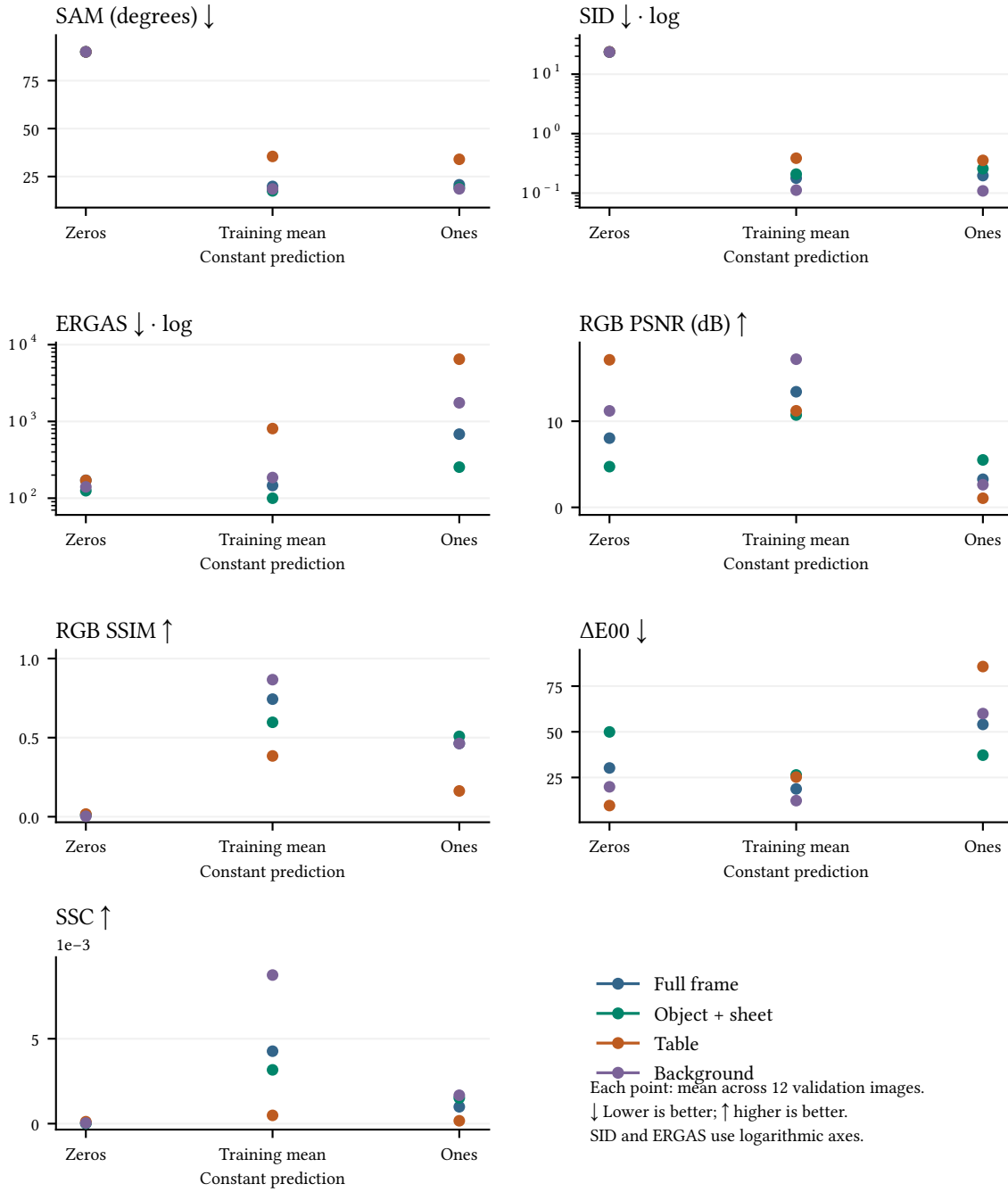


Figure 26: M9: Constant cubes. Six submetrics and SSC for each setting and region; points show equal-image means over the twelve validation images.

D How were the Raw2HSI comparisons run?

D.1 Fixed architecture and recipe

The organizer baseline packs each 2×2 mosaic block into four channels. A 64-channel feature extractor has a head, eight residual blocks and a tail, using convolutions, BatchNorm and SiLU. A learned output projection and a projected input skip produce 61×4 values per packed position; PixelShuffle arranges them into a full-resolution cube. Our comparisons retain this architecture. The RGB-input experiment adds interpolated color planes and 9840 input-layer parameters.

The main training objective is full-frame or masked mean L1 + 0.1 SAM in radians. Masked reductions average valid pixels within each image before averaging images. The training SAM offsets both input spectra by 10^{-8} , retains a denominator epsilon and clips the cosine away from the arccos endpoints, as specified in Appendix B. The loss clamp clips predictions to $[0, 1]$.

Training uses AdamW with initial learning rate 2×10^{-4} , weight decay 10^{-4} , betas 0.9 and 0.999, and epsilon 10^{-8} . The cosine schedule reaches 10^{-6} and advances after each epoch. Batch size is four. Every epoch traverses the 165 training images. There is no augmentation. Execution uses FP16 autocast, gradient scaling and compilation. The two long runs each record 13 overflow-skipped updates among 42,000 attempted steps.

Across the three paired 100-epoch seeds, masking reduces object recipe loss by 18.9%, 11.2% and 25.3%, averaging 18.4%. Object L1, SAM and ERGAS improve in each seed, while SID varies. Full-frame, table and background loss worsen in each seed.

Seed-0 screening compares full loss, removal of the training clamp, object + sheet loss, object + sheet + table loss and mosaic plus RGB input. The selection rule is final-epoch object + sheet loss at floor zero. It retains clamped training and object + sheet masking. Full, unclamped, masked and RGB-input arms then receive paired seeds 1 and 2. The long full and masked runs each use seed 0.

Arm	Seed	Epoch	Loss	SAM °	SID	ERGAS	SSC
Full	0	100	0.1002	12.1291	0.6826	33.7291	0.0061
Full	1	100	0.1052	13.5042	0.7483	34.0030	0.0054
Full	2	100	0.1041	13.0967	0.7195	36.5085	0.0053
Unclamped	0	100	0.1023	11.9837	0.1266	35.5271	0.0343
Unclamped	1	100	0.1170	13.4202	0.1675	36.0598	0.0244
Unclamped	2	100	0.1129	13.0474	0.1361	41.0986	0.0289
Masked	0	100	0.0813	10.7557	0.6791	27.7510	0.0105
Masked	1	100	0.0934	12.9615	0.7628	29.0751	0.0098
Masked	2	100	0.0778	9.0899	0.2873	24.8625	0.0279
Object + table	0	100	0.0942	13.1928	0.9016	34.4030	0.0072
RGB input	0	100	0.0956	12.0337	0.5707	34.7189	0.0068
RGB input	1	100	0.1007	13.2447	0.6652	36.4608	0.0058
RGB input	2	100	0.0979	12.2888	0.5645	36.3573	0.0060
Full	0	1000	0.0651	7.5236	0.1557	21.2122	0.0537
Masked	0	1000	0.0605	7.4604	0.2393	21.2599	0.0302

Table 4: Final scores of all fifteen local training runs, evaluated on object + sheet at floor zero. The sixteenth run uses the official split for the Kaggle reference.

D.2 Long-run readouts and score accounting

Masked training improves RGB PSNR, SSIM and $\Delta E00$ on all twelve object + sheet regions at epoch 1000, but worsens SID on all twelve. Mean SAM and ERGAS remain close, with six scenes favoring each model. The surrounding regions worsen more broadly. The following chart preserves every region and individual scene.

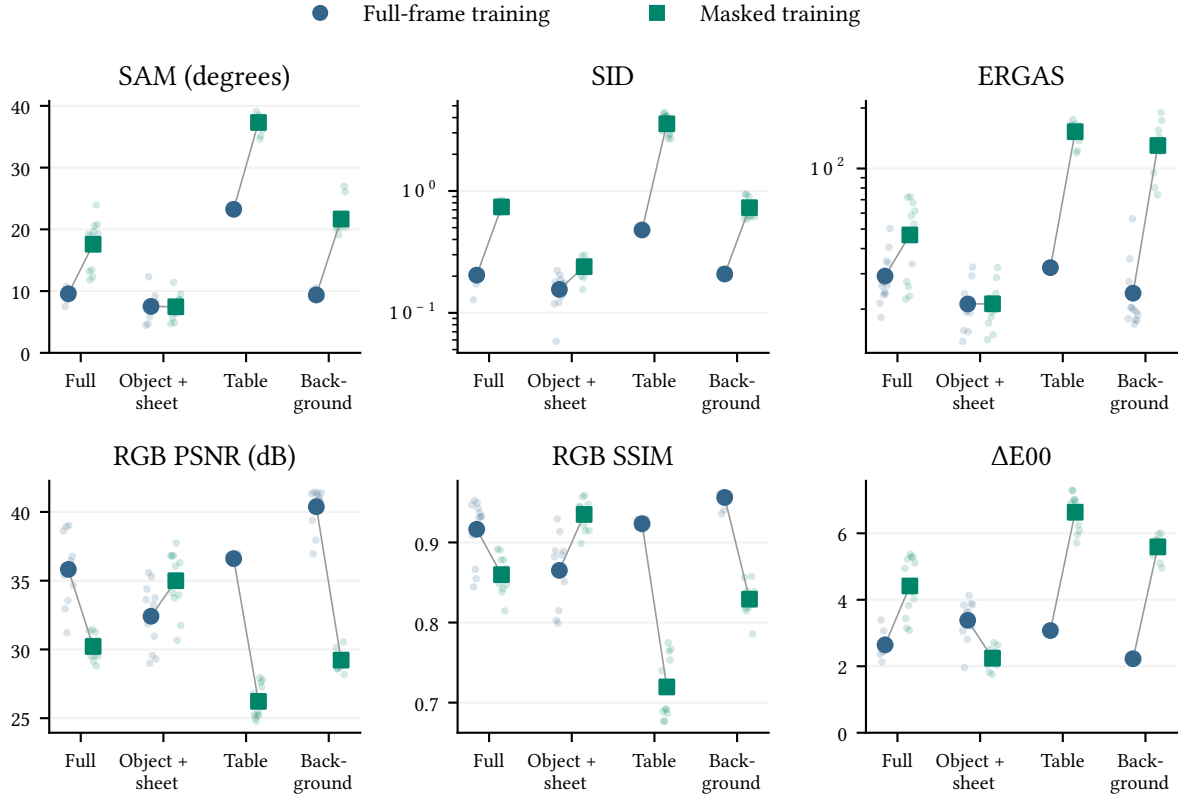


Figure 27: The six regional Raw2HSI metrics at epoch 1000 and floor zero. Small dots show individual scenes; large markers show means. SID and ERGAS use logarithmic scales. Lower is better for SAM, SID, ERGAS and $\Delta E00$; higher is better for PSNR and SSIM.

Object + sheet readout	Full-frame loss	Masked loss
L1 + 0.1 SAM	0.06507	0.06045
L1	0.05194	0.04743
SAM, degrees	7.5236	7.4604
SID	0.1557	0.2393
ERGAS	21.2122	21.2599
RGB PSNR, dB	32.4140	34.9985
RGB SSIM	0.8651	0.9349
$\Delta E00$	3.3819	2.2427
SSC	0.05372	0.03025

Table 5: Epoch-1000 object + sheet results at floor zero. Both models use seed 0 and the same 1000-epoch schedule.

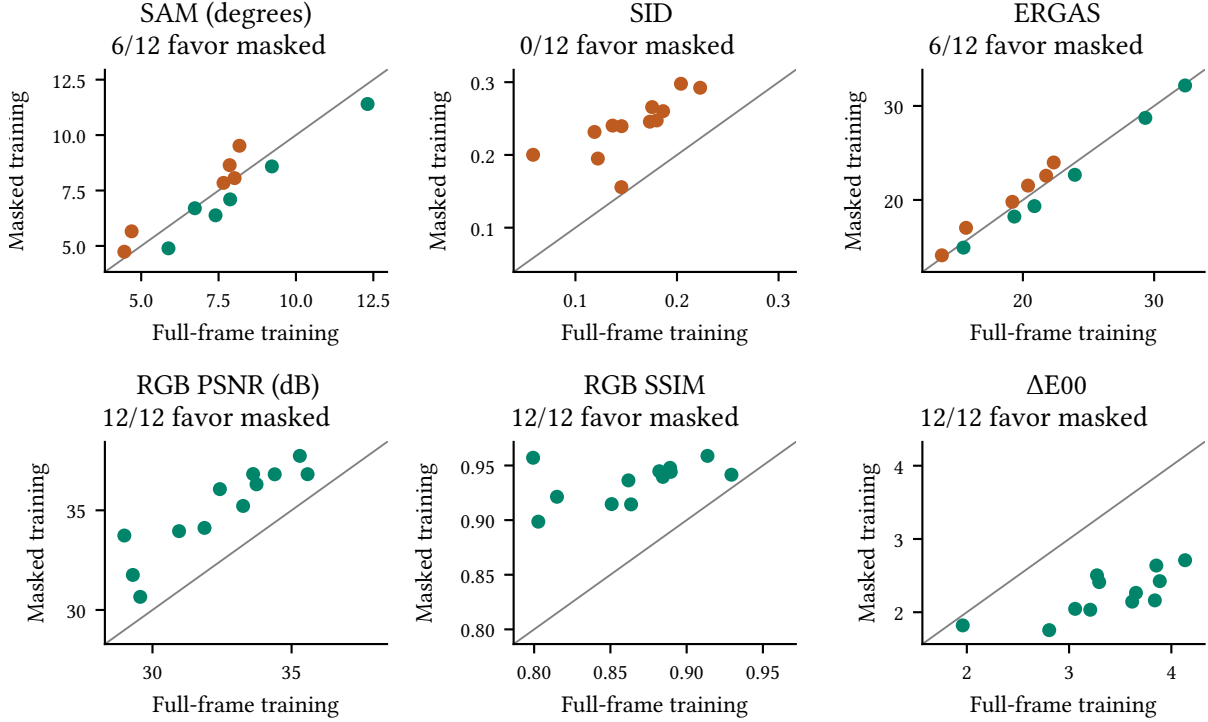


Figure 28: Raw2HSI object + sheet metrics at epoch 1000 and floor zero. Each point compares the same scene under full-frame and masked training. Green points favor masked training and orange points favor full-frame training. The diagonal indicates equality.

Model	Floor	L1	SID	SSC
Full-frame training	0	0.05194	0.15569	0.05372
Full-frame training	0.005	0.05190	0.04439	0.13039
Masked training	0	0.04743	0.23929	0.03025
Masked training	0.005	0.04737	0.05685	0.13113

Table 6: The same epoch-1000 predictions evaluated with two floors. The target is unchanged. Means over twelve object + sheet regions.

At floor zero, object + sheet SSC favors full-frame training, 0.05372 versus 0.03025. Flooring the same predictions at 0.005 gives nearly equal arithmetic means, 0.13039 and 0.13113. Six scenes favor each model and the median paired change remains negative. L1 changes little. SID is the largest component opposing masking; its contribution shrinks after flooring.

At epoch 1000, full-frame loss is 0.04031 for full training and 0.07236 for masked training; table loss is 0.04424 and 0.07925; background loss is 0.02271 and 0.07676. All twelve scenes worsen in those readouts under masking. Object + sheet loss improves on ten of twelve scenes.

For positive SSC, the exact accounting is

$$\log(\text{SSC}) = \frac{\log S_{\text{SAM}} + \log S_{\text{SID}} + \log S_{\text{ERGAS}}}{6} + 0.35 \log S_{\text{SPAT}} + 0.15 \log S_{\text{DE}}. \quad (11)$$

For each image, subtracting the full-model terms from the masked-model terms gives the log of their SSC ratio. This remains exact with component floors because it uses the transformed values. Averaging these terms gives the mean log of the paired score ratios.

Term	Floor 0	Floor 0.005
SAM	+0.0021	+0.0027
SID	−0.6706	−0.1038
ERGAS	−0.0026	−0.0009
Spatial	+0.0409	+0.0402
Color	+0.0570	+0.0557

Table 7: Mean contributions to $\log(\text{masked SSC} / \text{full SSC})$, calculated per image before averaging. Spatial combines PSNR and SSIM. These terms sum to a mean log-ratio.

Table 7 reports these five terms for both floors. Exponentiating a mean log contribution gives its multiplicative factor in the geometric mean of paired score ratios, holding the other terms fixed. For SID, $\exp(-0.6706) \approx 0.51$ at floor zero and $\exp(-0.1038) \approx 0.90$ at floor 0.005. Exponentiating the sum of all terms gives paired geometric mean ratios of 0.564 and 0.994. The latter still slightly favors full-frame training, whereas the arithmetic mean SSC slightly favors masking.

D.3 The repeating zero pattern

The phase audit examines each wavelength at each of the four positions in the repeated 2×2 output block. The following five wavelength/position pairs have negative raw predictions at that position throughout every object + sheet region at epoch 1000. Clamping turns those outputs into zero.

Training	Band, nm	2×2 position	Maximum raw	Mean target
Full	530	Top left	-0.2250	0.2973
Full	420	Bottom right	-0.4278	0.2478
Masked	430	Top left	-0.3559	0.2424
Masked	420	Bottom right	-0.2725	0.2478
Masked	680	Bottom right	-0.6282	0.4514

Table 8: Confirmed wavelength/position pairs. At each named position, every object + sheet raw prediction is negative across all twelve scenes. Target values are means of per-image means.

The full and masked models have partially zero spectra at 50.87% and 50.14% of object pixels respectively. These pixels account for 92.43% and 95.78% of SID, averaged as per-image shares. The worst 5% of pixels contribute about 17% of SID, so this problem extends across many pixels. More than 99.96% of object SID occurs where mean target reflectance exceeds 0.02.

The listed full-model wavelengths account for 37.2% of object ERGAS squared, and the masked-model wavelengths for 51.8%, as means of per-image band shares. Those shares include every error in each named band. They do not isolate the zero position alone. A clamp has zero derivative for raw outputs outside its range, but shared parameters can still change those outputs.

D.4 How component bounds affect a training trajectory

The SID component reaches its lower bound when SID exceeds approximately 0.2763. At epoch 100 within the long schedules, both models are saturated on all twelve object regions. Their differing SID values therefore make identical contributions to SSC. By epoch 1000, none of the full-frame model’s scenes remain saturated, while two masked-model scenes do.



Figure 29: The long Raw2HSI trajectories at floor zero, starting at epoch 100. Left: equal-image mean SID; the dashed line marks the component saturation threshold. Middle: the number of individually saturated scenes. Right: mean SSC.

Between epochs 100 and 1000, full-frame training reduces object loss by 27.3%, from 0.08956 to 0.06507; masked training reduces it by 22.1%, from 0.07757 to 0.06045. These are improvements within each long trajectory. The independently trained 100-epoch models have different learning-rate histories. The results show why the raw metrics must remain visible during training: a metric can improve while its bounded SSC component stays fixed, then affect the score after crossing the threshold. They leave the explanation of what scene structure the model learned unresolved.

D.5 Raw2HSI regional loss and prediction floors

At epoch 1000, mean object + sheet recipe loss is 0.06045 with masked training and 0.06507 with full-frame training. The surrounding regions worsen under masking. The following retained comparison shows how changing the prediction floor changes the SSC assessment of these same models.

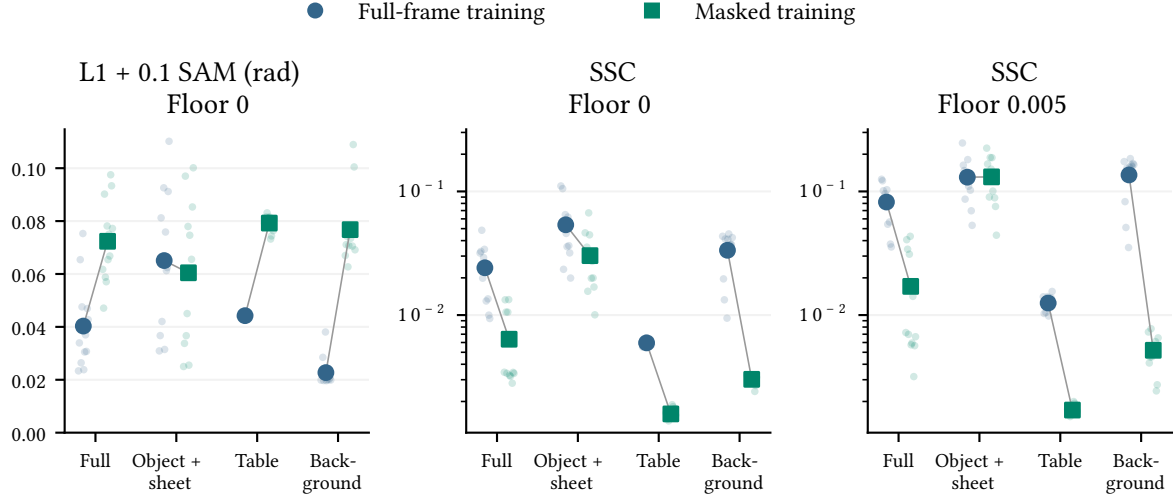


Figure 30: Full-frame and masked Raw2HSI training at epoch 1000, evaluated on twelve validation images across all four regions. Small dots show scenes; large markers show equal-image means. Left: L1 + 0.1 SAM in radians, lower is better. Middle: SSC after clipping predictions to $[0, 1]$. Right: SSC after raising the lower bound to 0.005, with the same upper bound 1 and unchanged targets. Both SSC panels use the same logarithmic scale.

E How was the deployed scorer tested?

This appendix examines the exact local calculation and the Kaggle SSC scores obtained in our experiments. The scores from these experiments concern the four private images from the challenge, shown as RGB renders in Figure 2. They do not reproduce the final hidden evaluation reported by the organizers.

E.1 Source reconstruction

The organizer-code snapshot corresponds to commit 927aafde57ca03c2909b34d885455d24d49fe4c4 of the public repository [7]. The normalized calculation was recovered from `utils/__pycache__/leaderboard_ssc.cpython-312.pyc`. The same checkout contains a different source implementation described as a toy scorer. The commit identifies the inspected public code, but Kaggle’s deployed version remains unknown.

The challenge overview paper describes a weighted arithmetic mean of the spectral, spatial and color groups, with PSNR calculated over the hyperspectral cube; its account does not mention SSIM [5]. The recovered calculation from bytecode instead combines the groups with a weighted geometric mean. Its spatial group includes SSIM and PSNR, both calculated on rendered RGB [7]. These are the specific source discrepancies behind our separate treatment of local measurements and deployed scores.

E.2 Exact local SSC combination

To form SSC, we convert the error metrics to component similarity scores, group them by spectral, spatial and color agreement, and combine the groups with a weighted geometric mean. For a raw metric value m , define

$$E_\tau(m) = \min\left(1, \max\left(10^{-6}, \exp\left(-\frac{m}{\tau}\right)\right)\right). \quad (12)$$

The spectral components use $\tau = 5$ for SAM in degrees, 0.02 for SID and 3 for ERGAS. The color component uses $\tau = 3$ for ΔE_{00} . PSNR is transformed as $S_{\text{PSNR}} = \min\left(1, \max\left(0, \frac{\text{PSNR} - 20}{30}\right)\right)$. The groups and final score are

$$\begin{aligned} S_{\text{spec}} &= (E_5(\text{SAM})E_{0.02}(\text{SID})E_3(\text{ERGAS}))^{\frac{1}{3}}, \\ S_{\text{spat}} &= \frac{S_{\text{PSNR}} + \text{SSIM}}{2}, \\ S_{\text{col}} &= E_3(\Delta E_{00}), \\ \text{SSC} &= S_{\text{spec}}^{0.5} S_{\text{spat}}^{0.35} S_{\text{col}}^{0.15}. \end{aligned} \quad (13)$$

SSIM enters the spatial group without an additional exponential transformation. Once a component reaches its lower bound, further deterioration of its raw metric does not change that component’s contribution to SSC.

E.3 Constant predictions as controls

Constant cubes ignore the input and scene structure. Full-frame local SSC averages are 2.94×10^{-5} for zeros, about 0.004 for the training mean spectrum, and 9.92×10^{-4} for ones. Individual metrics can still look favorable: the training-mean cube receives background RGB SSIM 0.867, alongside ERGAS 185.81 and ΔE_{00} 12.32.

E.4 The fixed reference and controlled changes

To test the deployed evaluator, we submitted controlled changes to one fixed prediction archive. The reference came from a separate Raw2HSI run trained on the official 167-image partition. Its epoch-992

checkpoint was selected by full-frame loss on the eleven public images. The archive contains predictions for the four private mosaics used in our Kaggle probes.

For the regional probes, we selected 16,384 pixels in each private image, equal to 1.5625% of its area. We set all 61 band values there to one and preserved every value elsewhere. Background and table probes therefore left object + sheet predictions unchanged. The object probe supplied an equal-area control. Figure 31 shows the selections on the input mosaics.

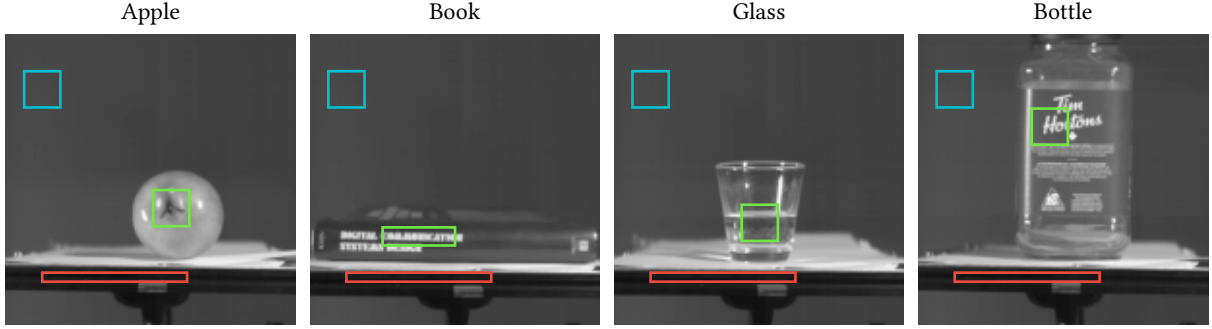


Figure 31: The four private input mosaics and probe rectangles. Cyan marks background, red marks table, and green marks object. The display is brightened for visibility.

Submission	Kaggle SSC
Unchanged reference replay	0.27299
Background patch set to one	0.15035
Table patch set to one	0.15028
Object patch set to one	0.16595
All values zero	0.13970
All values 10^{-6}	0.14094
710–1000 nm set to zero	0.14342
710–1000 nm alternating zero/one	0.11265

Table 9: Eight controlled submissions from one reference archive. Kaggle returned aggregate scores; per-image and per-metric results are unavailable.

The background and table changes reduced SSC by about 45%; the object change reduced it by about 39%. This directly establishes that the selected surrounding regions influence the deployed evaluation. The result does not identify which metrics carry that influence or whether some metrics use masks. Equal area also leaves brightness, spectra and patch perimeter unequal, so the percentages do not establish an intrinsic ordering of region importance.

E.5 A zero prediction contradicts the complete local candidate

A Kaggle submission of all-zero cubes scored 0.13970. Under the local SAM convention, an exactly zero prediction receives 90° at every evaluated pixel. Its transformed SAM component reaches 10^{-6} . SAM has exponent $\frac{0.5}{3} = \frac{1}{6}$ in the final geometric score. With every other bounded factor at its maximum,

$$\text{SSC} \leq (10^{-6})^{\frac{1}{6}} = 0.1. \quad (14)$$

The same upper bound applies after averaging image scores and for any nonempty spatial mask under this zero convention.

Kaggle’s 0.13970 exceeds that bound. Different private target images cannot explain the contradiction with this complete combination of formula, weights and zero handling. The bound is deliberately generous: the local zero controls actually score much lower because the other factors are also poor. Changing zeros to 10^{-6} raises Kaggle’s score only to 0.14094.

E.6 Near-infrared changes affect the deployed score

The two near-infrared probes preserve every 400–700 nm value. They also produce exactly the same RGB arrays as the reference under our visible-band renderer, yet their Kaggle scores fall to 0.14342 and 0.11265. The modified bands therefore influence the deployed evaluation. SAM, SID and ERGAS can all respond to these changes; the returned aggregate cannot identify their respective contributions. Kaggle could also use a different spatial or color calculation.

E.7 What remains unresolved

The overview paper’s arithmetic combination and hyperspectral PSNR differ from the recovered code’s geometric combination and RGB PSNR/SSIM. Changing only the outer mean in local calculations produces very different values, but comparisons across different image sets cannot select the deployed formula. We reject the recovered candidate as a complete account of the service while leaving its actual transformations, weights, masks and aggregation unresolved. The observed constant scores provide no universal lower bound for the leaderboard.

F Additional model settings and result records

F.1 Patch training and evaluation

The fourteen runs in Section 5 use the same 165/12 split and frozen masks. Four distinct training images are selected uniformly per update, then an eligible origin uniformly within each image. Origins lie on a four-pixel grid from 0 to 896. Every image has crops meeting the masked policy’s 50% object + sheet threshold. Loss averages valid pixels within each patch, then the four patch means equally.

AdamW uses initial learning rate 2×10^{-4} , weight decay 10^{-4} , betas 0.9 and 0.999, and epsilon 10^{-8} . A cosine schedule advances after each reference epoch and reaches 10^{-6} at the chosen budget. All models start from seed 1. The common L1 + 0.1 SAM loss clamps predictions to $[0, 1]$ and retains the training-SAM stabilization in Appendix B. Execution uses FP32 with TF32 disabled, channels-last storage and CUDA-graph forward/backward operations. No gradient clipping, accumulation or activation checkpointing is used.

Detailed evaluation occurs before training, after epoch 1 and every five epochs. Each complete-image prediction supplies both floors and all regions. Main comparisons use final weights.

F.2 Reconstruction interfaces

All four configurations use base feature width 64. Raw MST++ has three distinct stages with widths 64, 128 and 256 through each encoder, reversed in the decoder. The packed variant starts from four measured RGGB planes and upsamples both the processed and shallow-feature branches. MSFN and SPECAT use whole-image Menon RGB with refinement before patch extraction, without extra clipping, gamma conversion or normalization.

MSFN links spectral and spatial encoder-decoders. SPECAT uses two downsampling steps and cumulative-attention block counts 2/2/1/2/2. Each example has its own RGB-derived guidance, and shifted attention applies its boundary mask. The outer residual addition is at width 64; a fixed selection of channels 0–60 supplies the target bands.

F.3 The IS-loss convention

Each SPECAT IS run uses the same seed-1 initialization and exact crop sequence as its original-loss parent under the same policy. For a clamped predicted spectrum p and target t , we use

$$\text{NSE}(p, t) = \frac{\sum_{b=1}^{61} |p_b - t_b|}{\max\left(\sum_{b=1}^{61} |t_b|, 10^{-8}\right)}, \quad (15)$$

$$L_{\text{IS}} = 10\text{L1} + \frac{\text{SAM}_{\text{rad}}}{3} + \text{NSE}. \quad (16)$$

NSE is calculated per valid pixel, then averaged within a patch and across patches. SAM uses radians and NSE uses a fraction. SATE prints NSE as a percentage but does not make its training-code units available [11].